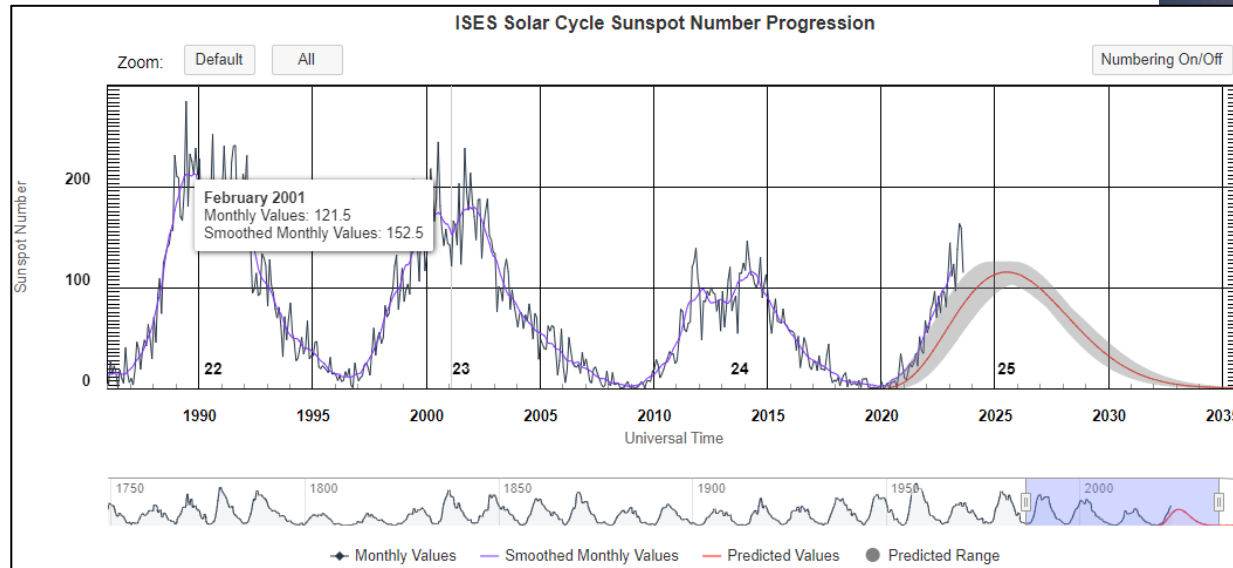


ASEN 5090 Lecture 08

Overview of Measurement Errors Ionospheric Effects



Penina.Axelrad@Colorado.edu

Thanks to Jason Li for expanded material in this lecture!

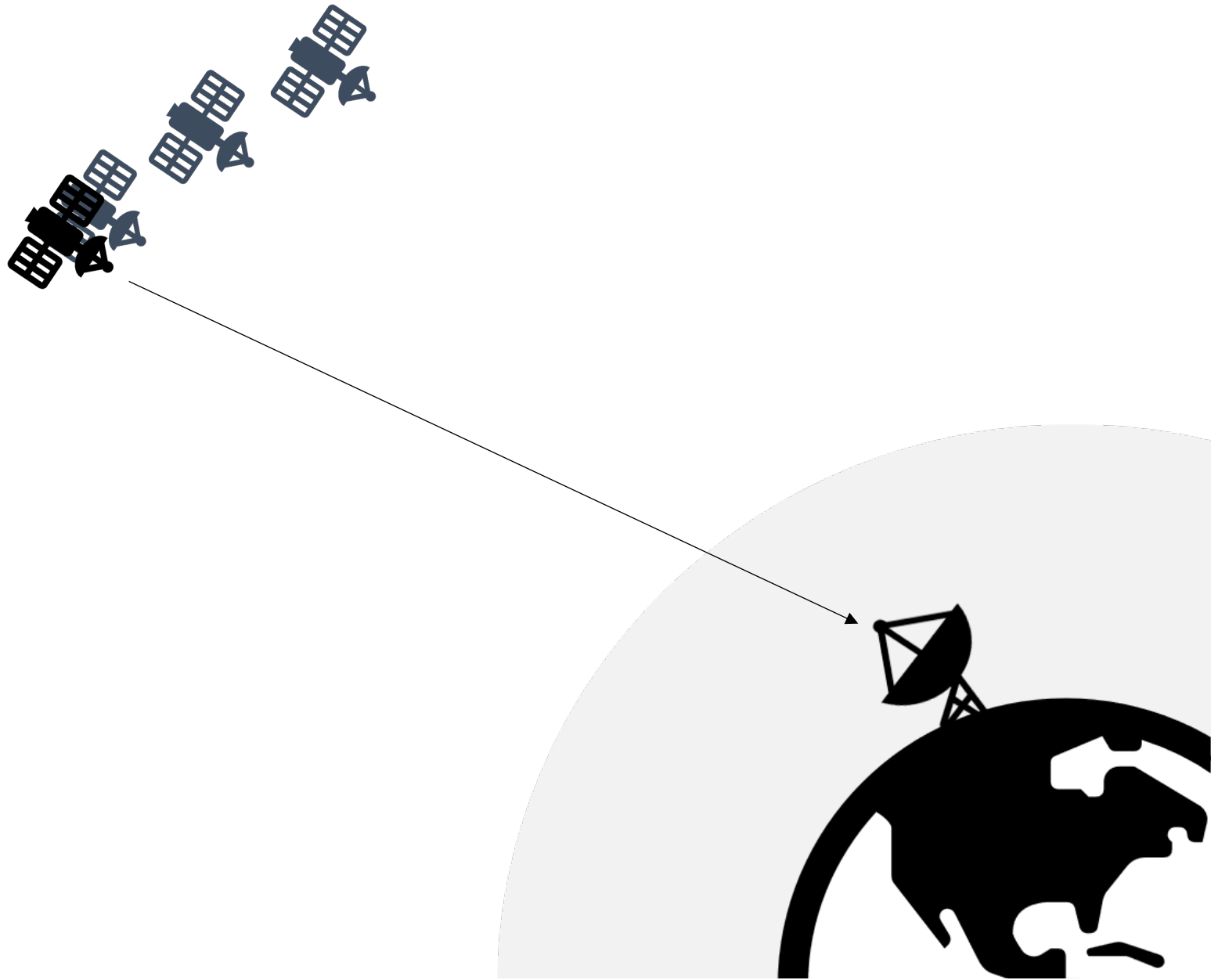
Pseudorange Model

$$\rho(t) = r(t, t - \tau) + c(\delta t_u(t) - \delta t^s(t - \tau)) + I_\rho(t) + T_\rho(t) + M_\rho(t) + \varepsilon_\rho(t)$$

- **GEOMETRIC RANGE**
- **CLOCKS**
- **RELATIVITY (not shown explicitly in the equation above)**
- **IONOSPHERE (FREE ELECTRONS)**
- **TROPOSPHERE (NEUTRAL ATMOSPHERE)**
- **MULTIPATH**
- **MEASUREMENT NOISE**

GEOMETRIC RANGE

- 1) Compute the GPS satellite position in ECEF at T_r based on the ***broadcast ephemeris*** or final orbits (sp3).
- 2) Use an a priori value for the receiver coordinates (RX) to find geometric range $R = |r_{GPS} - r_{RX}|$.
- 3) Compute the time of transmission: $T_t = T_r - R/c$ (use the real speed of light, NOT $3e8$)
- 4) Compute the satellite position at T_t in ECEF at T_t based on the broadcast ephemeris.
- 5) Rotate the satellite position to ECEF at T_r .
- 6) Compute a new geometric range using this position for r_{GPS} , $R = |r_{GPS} - r_{RX}|$.
- 7) Repeat steps 3-6 until convergence. (If it takes more than two iterations, something is wrong).



Error Models

- Noise vs. bias
 - Noise: quickly varying random errors that average to zero
 - Bias: Errors that are persistent, cannot be averaged out
- Three categories of GPS errors:
 - Errors that can be corrected through control segment uploaded parameters that are broadcast in the navigation message:
SV clock, orbit, ...
 - Errors associated with the propagation medium: iono, tropo,...
 - Receiver biases, noise and interference near the receiver
(multipath)

Control Segment Errors: SV Clock and Ephemeris

- Satellites broadcast predicted orbital parameters & clock error model parameters in navigation message
- How are these models computed?
 - Ground-based monitor station measurements
 - Kalman filter algorithms compute current SV position, velocity, and clock errors
 - Prediction model generates ephemeris and clock parameters to be uploaded to satellites
 - Satellites broadcast the parameters in the navigation message
- How reliable are these predictions & corrections?
 - There are errors in both current values and predicted values of parameters
 - Errors grow with time
 - Ephemeris and clock error RMS specification: $\sim 3\text{m}$



SIS SPS Specification

- <https://www.gps.gov/gps-performance>

mode, one or more satellites are not uploaded with fresh NAV message data by the Control Segment on a regular basis. See paragraph A.4.3.2 for additional information.

3.4.1 SPS SIS URE Accuracy Standards

The SPS SIS URE accuracy shall be as specified in Table 3.4-1.

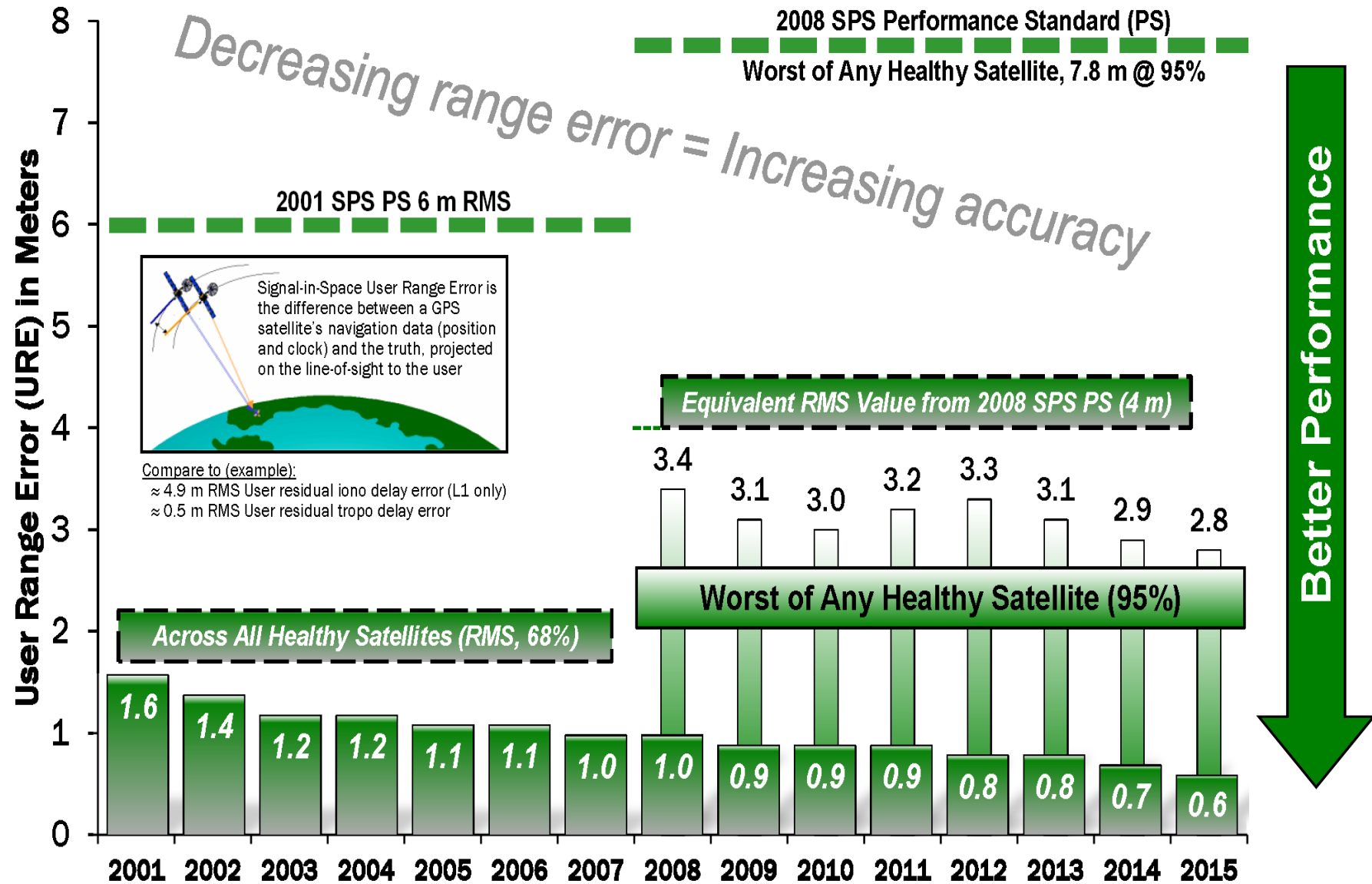
Table 3.4-1. SPS SIS URE Accuracy Standards

SIS Accuracy Standard	Conditions and Constraints
Each SPS SIS Component Combination per Table 2.2-2: • ≤ 7.0 m 95% Global Statistic URE during Normal Operations over all AODs • ≤ 3.8 m 95% Global Statistic URE during Normal Operations at Zero AOD • ≤ 9.7 m 95% Global Statistic URE during Normal Operations at Any AOD	• For any trackable and healthy SPS SIS • Neglecting SF ionospheric delay model errors • Including group delay time correction (T_{GD}) errors at L1 • Including inter-signal bias (P(Y)-code to C/A-code) errors at L1 • Including ISC errors
Each SPS SIS Component Combination per Table 2.2-2: • ≤ 30 m 99.94% Global Statistic URE during Normal Operations • ≤ 30 m 99.79% Worst Case Single Point Statistic URE during Normal Operations	• For any trackable and healthy SPS SIS • Neglecting SF ionospheric delay model errors • Including group delay time correction (T_{GD}) errors at L1 • Including inter-signal bias (P(Y)-code to C/A-code) errors at L1 • Including ISC errors • Standard based on measurement interval of one year; statistic of daily values within the service volume • Standard based on 3 service failures per year, lasting no more than 6 hours each
SF C/A-Code: • ≤ 388 m 95% Global Statistic URE during Extended Operations after 14 Days without Upload	• For any trackable and healthy SPS SIS
Each SPS SIS Component Combination per Table 2.2-2: • ≤ 2.0 m 95% Global Statistic URE during Normal Operations over all AODs for the ensemble of constellation slots	• Across all trackable and healthy SPS SISs from satellites occupying constellation slots • Neglecting SF ionospheric delay model errors • Including group delay time correction (T_{GD}) errors at L1 • Including inter-signal bias (P(Y)-code to C/A-code) errors at L1



Standard Positioning Service (SPS) Signal-in-space ranging errors (SISRE) history from Aerospace Corporation

Standard Positioning Service (SPS) Signal-in-Space Performance



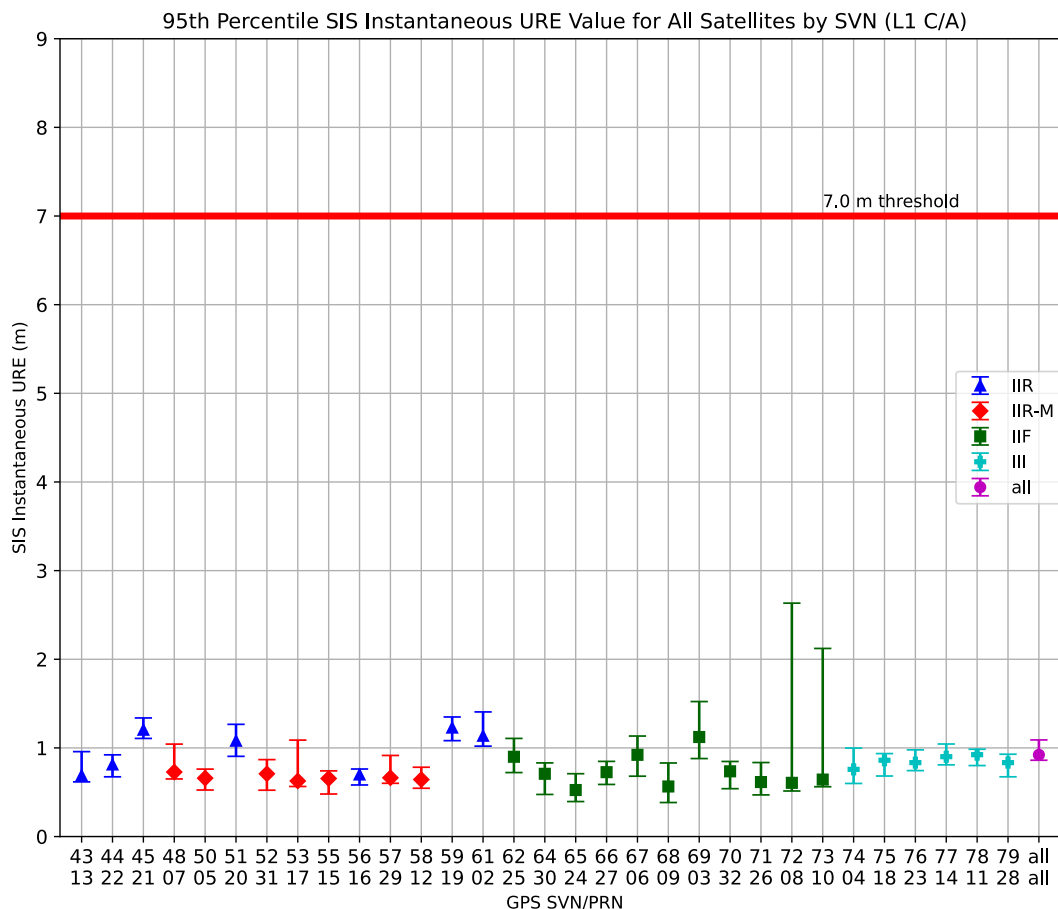
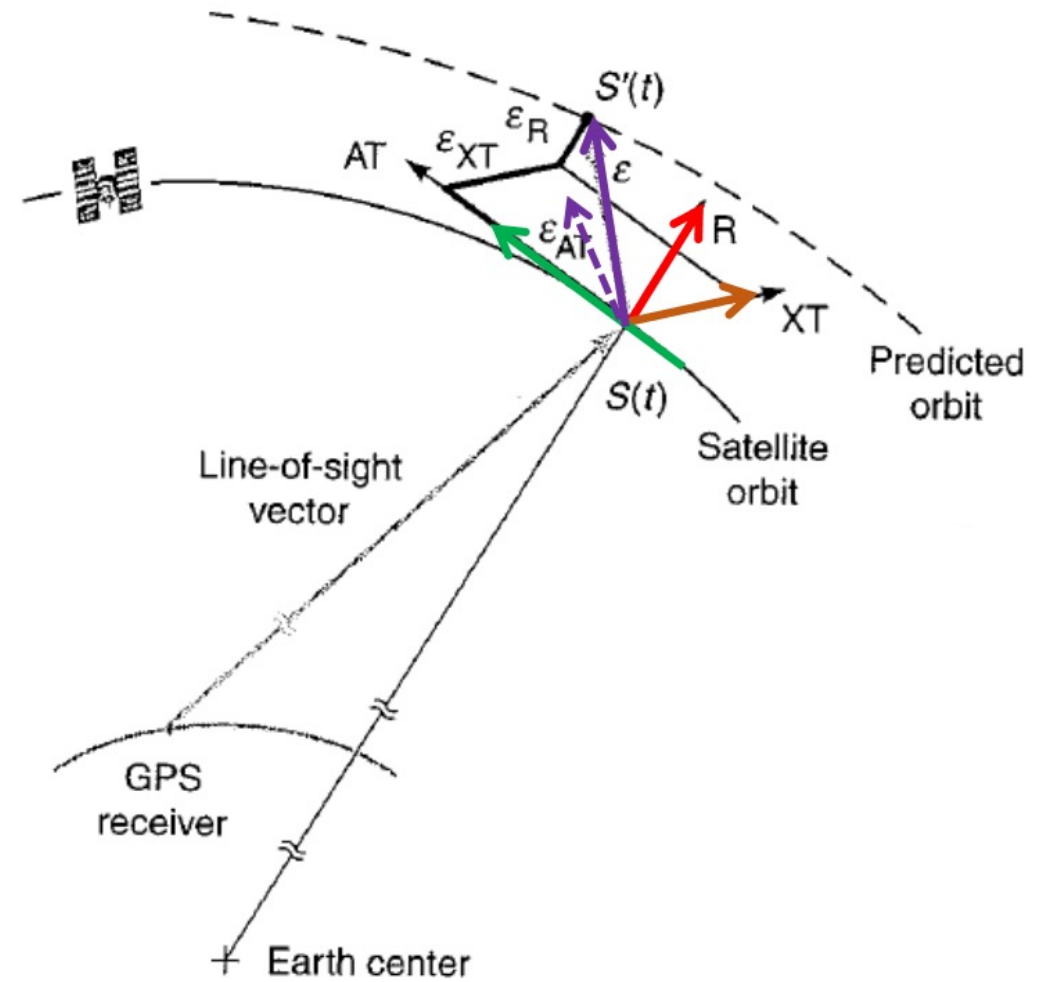
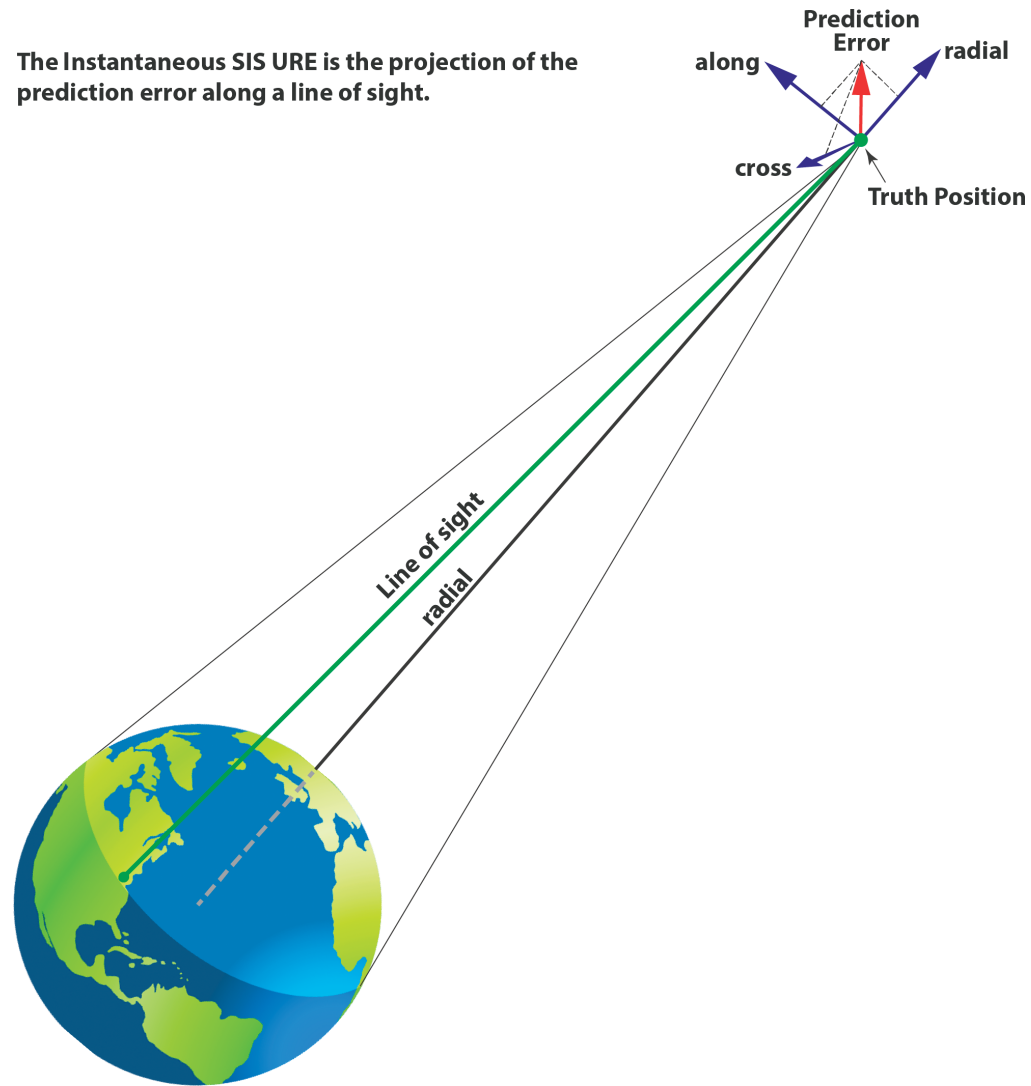


Figure 3.1: Range of the L1 C/A Monthly 95th Percentile Values for All SVs

Notes: Each SVN with valid data is shown sequentially along the horizontal axis. The median value of the monthly 95th percentile SIS URE is displayed as a point on each of the vertical bars. The minimum and maximum of the monthly 95th percentile SIS URE for 2024 are shown by whiskers at the top and bottom of each vertical bar. Color and shape distinguish between the Block IIR, Block IIR-M, Block IIF, and GPS III SVs. The red horizontal line at 7.0 m indicates the upper bound given by the SPS PS 2020 Section 3.4 performance metric. The marker for “all” represents the monthly 95th percentile values across all satellites.

Ephemeris Error Components



Textbook, P156, Fig. 5.5



Signal in space range error (SISRE)

Global average range error due to GPS orbit & clock errors:

$$s = \sqrt{w_1^2(\Delta r_R - \Delta T)^2 + w_2^2(\Delta r_A^2 + \Delta r_C^2)}$$

Where R = radial, A = along-track, C = cross-track

For GPS w/2 deg elevation cutoff

$$w_1 = 0.980, \quad w_2 = 0.141$$

GPS Solutions (2025) 29:33
<https://doi.org/10.1007/s10291-024-01793-6>

ORIGINAL ARTICLE

The 2024 GPS accuracy improvement initiative

O. Montenbruck¹ · P. Steigenberger¹

<https://link.springer.com/article/10.1007/s10291-024-01793-6>

<https://doi.org/10.1016/j.asr.2018.03.041>

3022

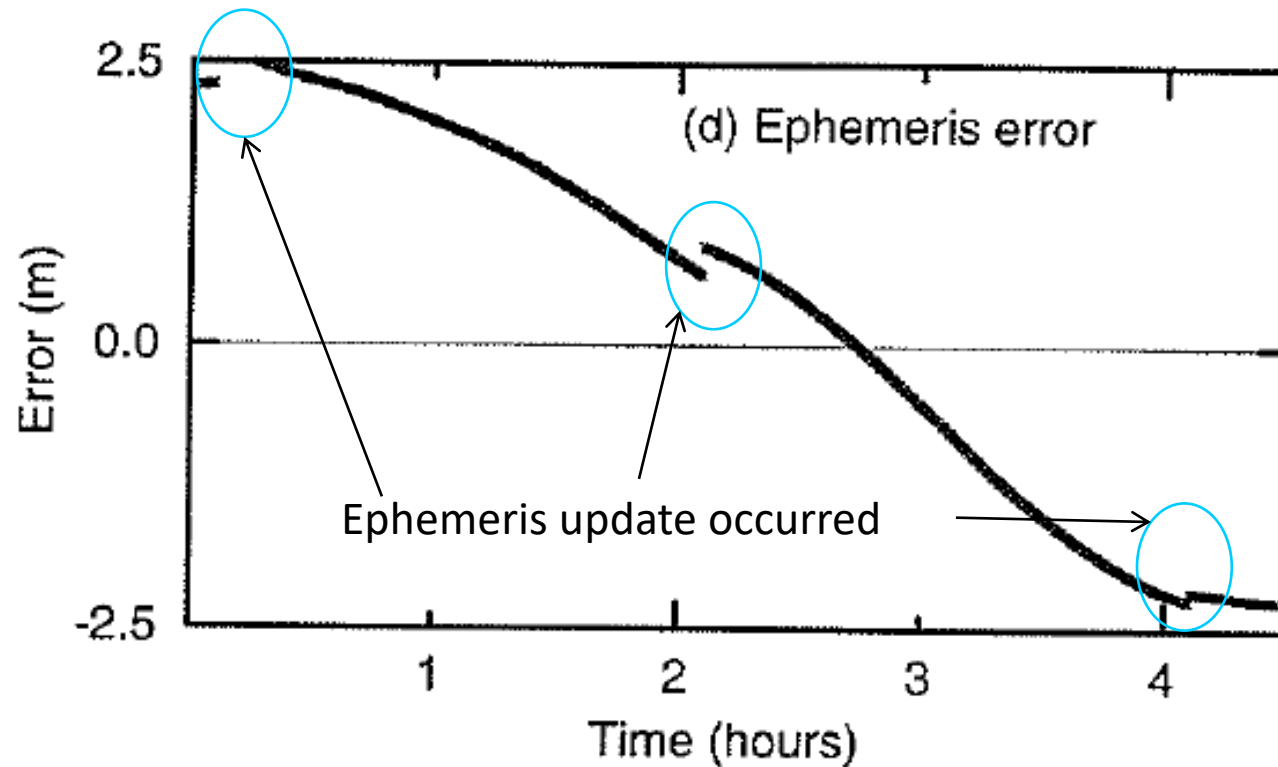
O. Montenbruck et al. / Advances in Space Research 61 (2018) 3020–3038

Table 1
SISRE performance specifications for the publicly accessible navigation services (Open Service (OS), Standard Positioning Service (SPS), and Positioning, Navigation, and Timing (PNT) service) of current global and regional navigation satellite systems.

Constellation (Service)	Specification	Notes
GPS (SPS)	≤ 7.8 m 95 % global average URE during normal operations over all ages of data (AODs) for any healthy SPS SIS (DOD, 2008)	– Applies for single frequency C/A-code, excluding single-frequency ionospheric delay model errors but including group delay time correction and inter-signal bias errors – The signal-in-space (SIS) user range error (URE) includes the pseudorange error budget components assigned to the GPS space and control segments – The AOD denotes the time elapsed since generation of the orbit and clock information by the control segment – The global average URE denotes the RMS of the instantaneous URE over the specified coverage – Additional specifications apply for global average UREs at zero AOD and at any AOD, as well as the probabilities of not exceeding specific global average and worst case single point averaged URE limits during normal operations
Galileo (Initial OS)	≤ 7 m 95 % global average SIS ranging error over all AODs for any healthy SIS above 5° elevation and calculated over a 30-days period (EU, 2016b) ≤ 2 m SIS ranging accuracy on average over the constellation of the 95 % SIS ranging accuracy of each healthy SIS (EU, 2016b)	– The same limit applies for single-frequency (SF) signals and for each dual-frequency (DF) combination – The SF specification excludes ionosphere model errors
BeiDou-2 (OS)	≤ 2.5 m 95 % URE for any healthy satellite (CSNO, 2013)	– The same limit applies for single-frequency (SF) signals and for each dual-frequency (DF) combination
QZSS (PNT)	≤ 2.6 m 95 % SIS URE (Cabinet Office, 2017a)	– The specification does not distinguish between SF and DF signals – For SF signals the ionosphere model errors are neglected – Timing group delay errors are not addressed by the specification and should therefore be included into the URE budget – The performance standard does not describe, whether the URE budget applies for individual sites, as an average over the service area, or as a global average over the visible surface of the Earth
		– The specification applies for the L1 C/A, L1C, L2C, L5 signals of the PNT service but does not distinguish between SF and DF signals – Timing group delay errors are not addressed by the specification and should therefore be included into the URE budget – The performance standard does not describe, whether the URE budget applies for individual sites, as an average over the service area, or as a global average over the visible surface of the Earth

Empirical Data: SV Ephemeris Error (Orbit, Clock)

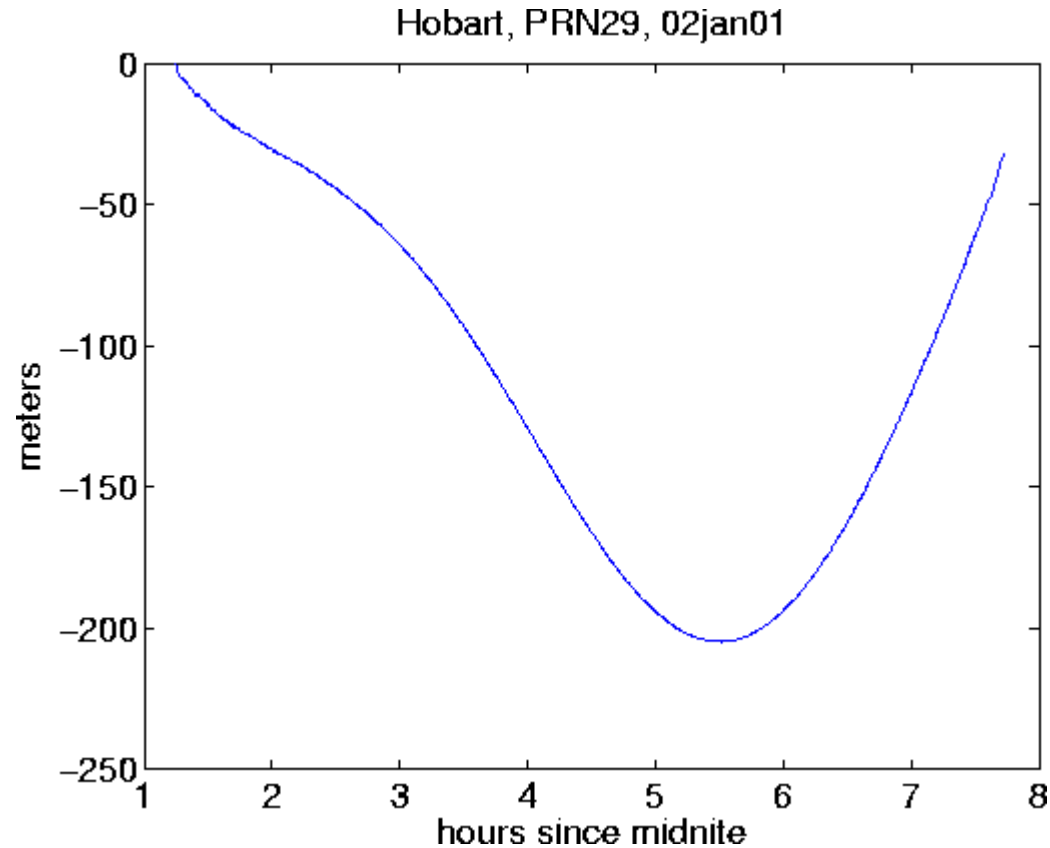
- Ephemeris error in computed range based on information in the broadcast message.
- Compare to precise ephemerides are available from IGS online
- Plot is based on difference between range computed using broadcast and IGS sp3



PSEUDORANGE minus GEOMETRIC RANGE

$$\rho_1 - R = c(\delta t_u - \delta t^s) + T + I_{\rho 1} + M_{\rho 1} + \epsilon_{\rho 1}$$

- Difference is typically dominated by receiver clock or satellite clock.
- How can you tell which?



CLOCKS and RELATIVITY

- Broadcast ephemeris

$$\Delta t^s = a_0 + a_1(t - toe) + a_2(t - toe)^2 + \Delta t_{RELATIVITY}$$

$$\Delta t_{RELATIVITY} = -\frac{2}{c^2} \sqrt{aGM} \cdot e \cdot \sin E$$

Notice this has a negative sign to be consistent with GPS interface document implementation (Relativity lecture material has a + sign)

IS-GPS-200

The user shall correct the time received from the SV with the equation (in seconds)

$$t = t_{sv} - \Delta t_{sv} \quad (1)$$

where

t = GPS system time (seconds),

t_{sv} = effective SV PRN code phase time at message transmission time (seconds),

Δt_{sv} = SV PRN code phase time offset (seconds).

The SV PRN code phase offset is given by

$$\Delta t_{sv} = a_{f0} + a_{f1}(t - t_{oc}) + a_{f2}(t - t_{oc})^2 + \Delta t_r \quad (2)$$

where

a_{f0} , a_{f1} and a_{f2} are the polynomial coefficients given in subframe 1, t_{oc} is the clock data reference time in seconds (reference paragraph 20.3.4.5), and Δt_r is the relativistic correction term (seconds) which is given by

$$\Delta t_r = F e \sqrt{A} \sin E_k.$$

The orbit parameters (e , $\sqrt{A}$, E_k) used here are described in discussions of data contained in subframes 2 and 3, while F is a constant whose value is

$$F = \frac{-2\sqrt{\mu}}{c^2} = -4.442807633 (10)^{-10} \frac{\text{sec}}{\sqrt{\text{meter}}},$$

Ionospheric Effects, Models, Corrections, and Observations

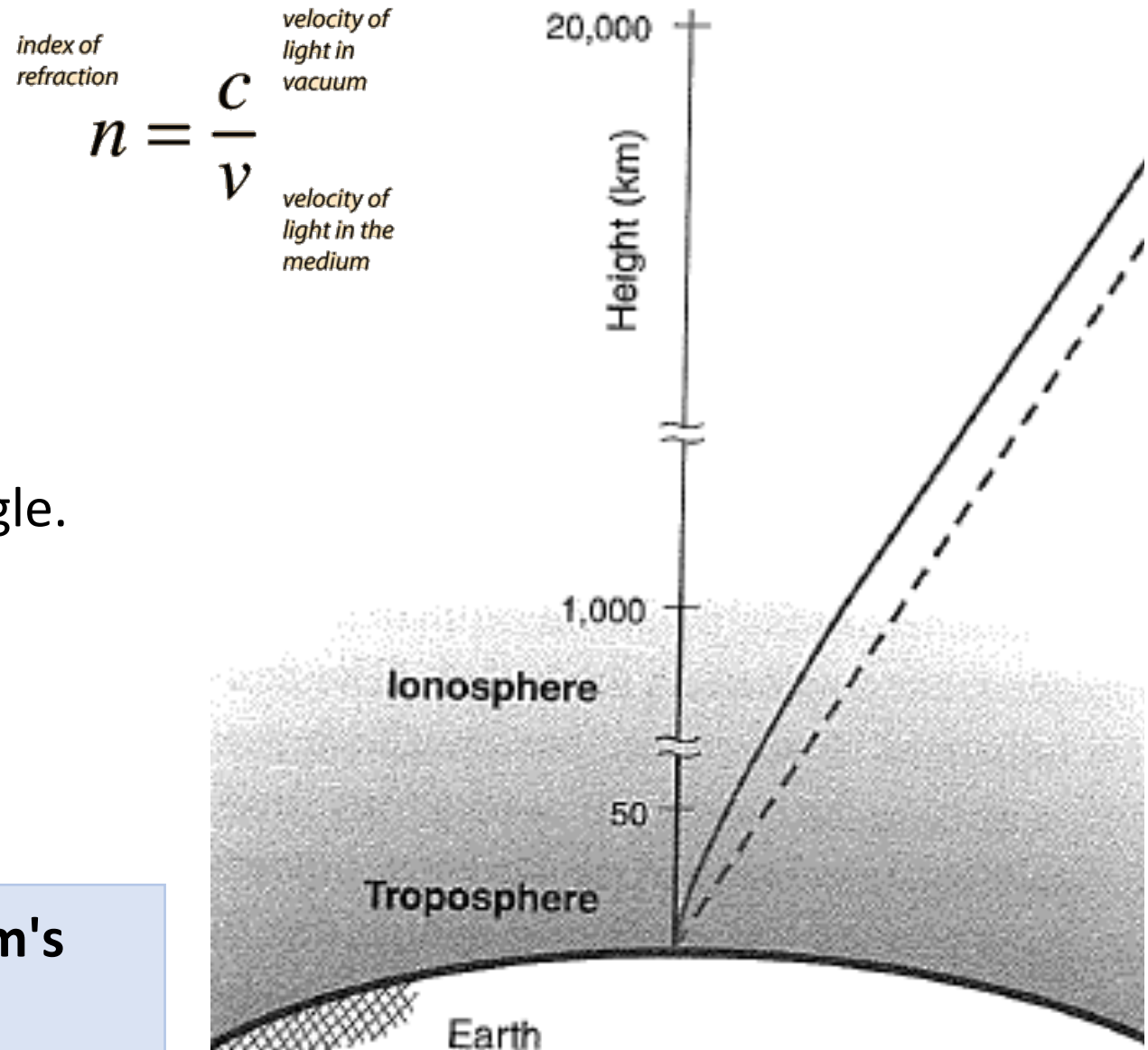
- 
- **What is it?** Introduction to ionosphere
 - **Why does it impact GNSS?** Group delay and phase advance
 - **What can we do about it?** Ionospheric modeling, correction, and mapping
 - **Ionospheric scintillation**
 - **Other ionospheric effects**

GPS Signal Propagation

- GPS signals are refracted in the Earth's atmosphere
- Bending is small for most ray paths
- Change in speed of propagation affects pseudorange and phase measurements
- Has a dependence on the incidence angle.
- Net effect as a delay:

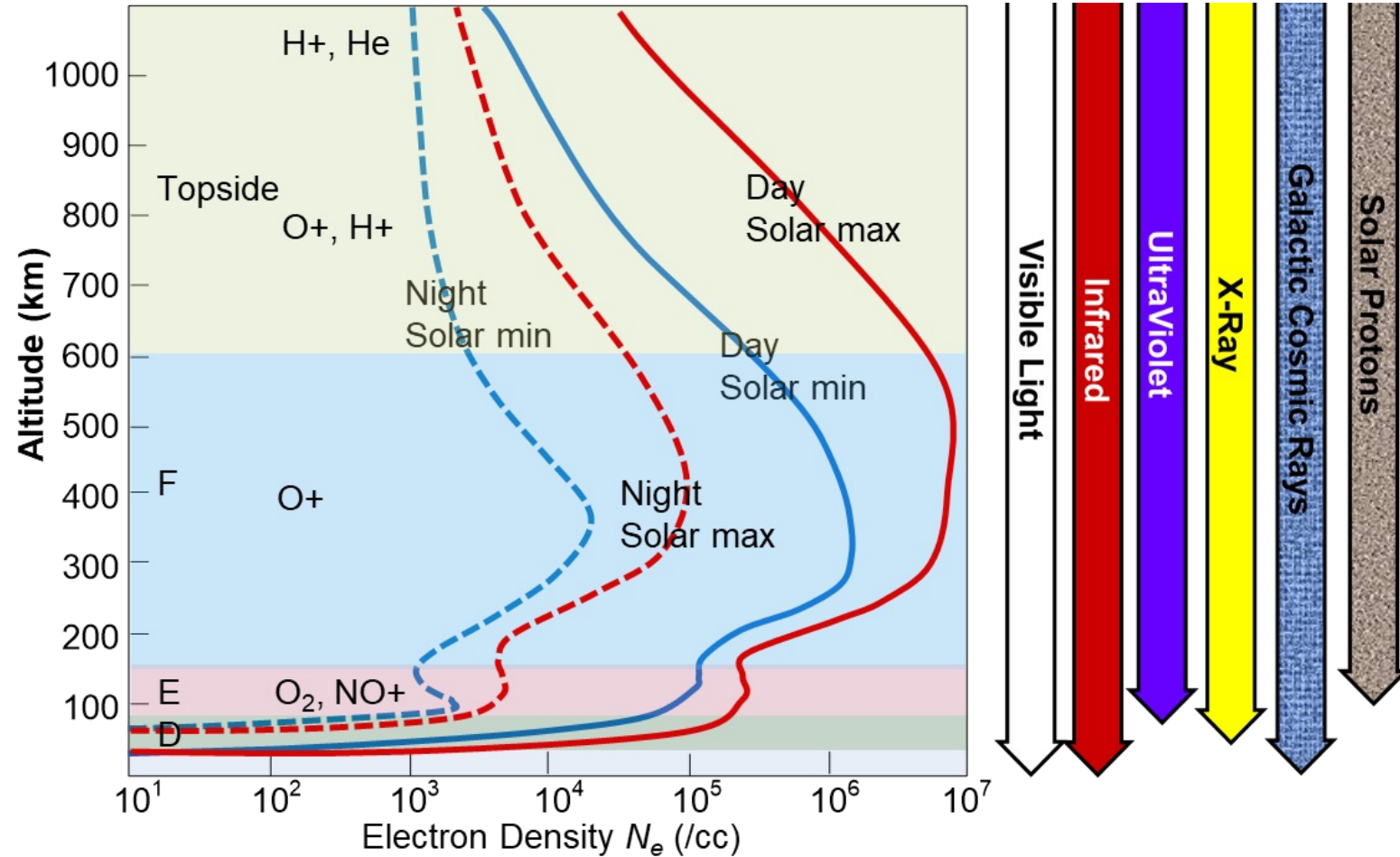
$$\Delta\tau = \tau - \tau_{vacuum} = \frac{1}{c} \int_s^R (n(l) - 1) dl$$

Speed of light is dependent on medium's refractive index.



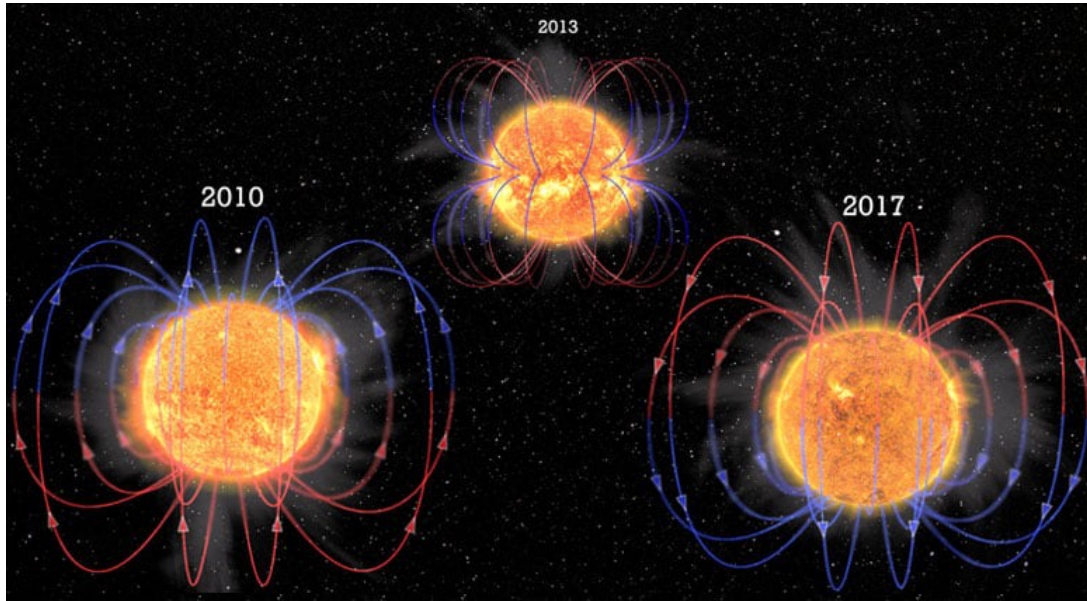
Introduction to Ionosphere

- Ionosphere is the region of the Earth's atmosphere, characterized as a weakly ionized plasma, from ~50km to 1000km height.
- Caused by solar radiation (EUV, X-ray) which partially ionizes the upper atmosphere
- Affects radio wave propagation: **refraction**, **diffraction**, reflection, Faraday rotation, absorption. The effect depends on the radio wave frequency.



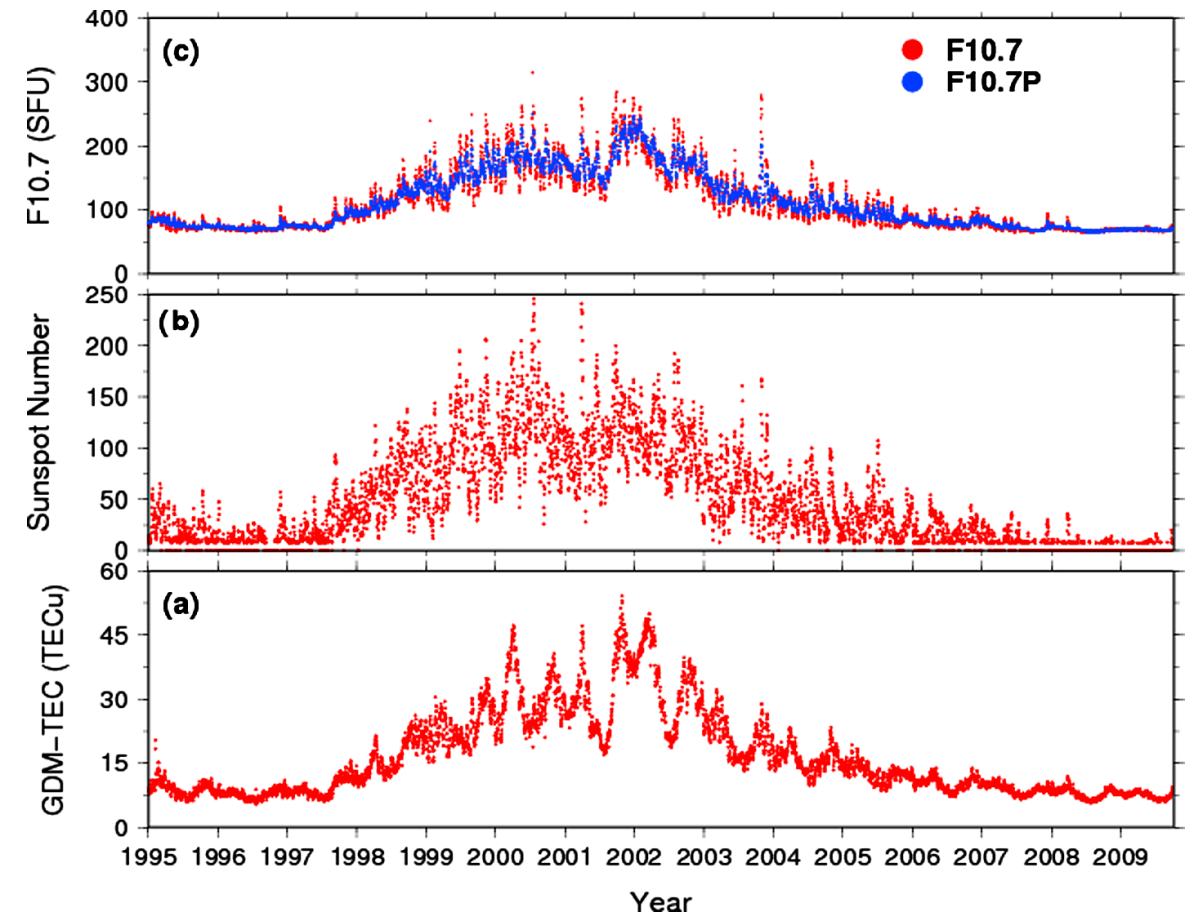
Ionospheric electron density is a function of altitude, time of day, season, etc...

Solar Cycle Influence



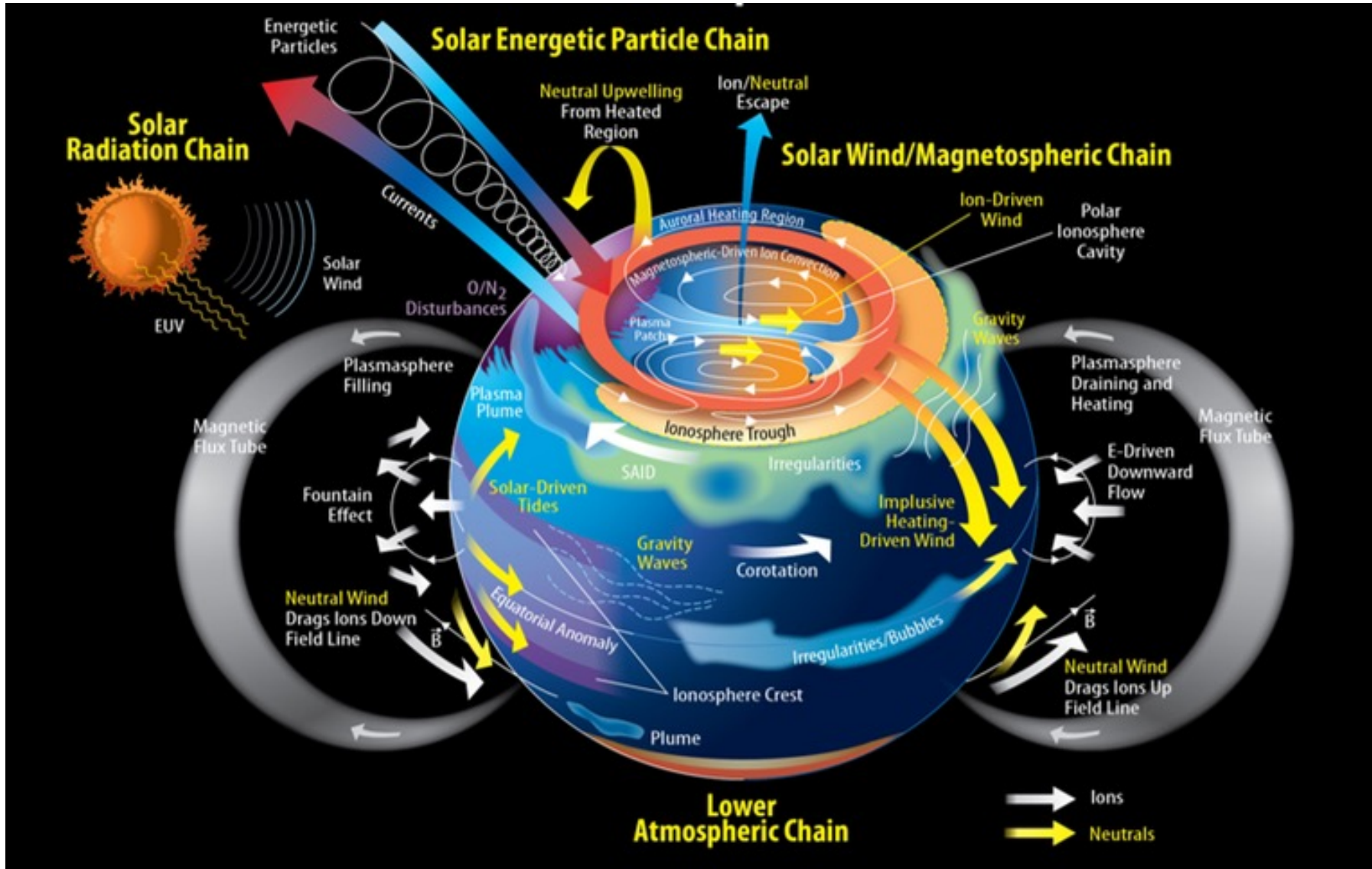
From:
Left: L. Almeida, Federal University of Rio Grande do Norte (UFRN), Brazil

Right: The influence of space weather on ionospheric total electron content during the 23rd solar cycle, Nicolas Bergeot, Ioanna Tsagouri, Carine Bruyninx, Juliette Legrand, Jean-Marie Chevalier, Pascale Defraigne, Quentin Baire and Eric Pottiaux
J. Space Weather Space Clim., 3 (2013) A25
DOI: <https://doi.org/10.1051/swsc/2013047x>



**Electron density increases during solar cycle maximum
which repeats every ~11 years**

Space Weather is Weather! Other Influences on Electron Density



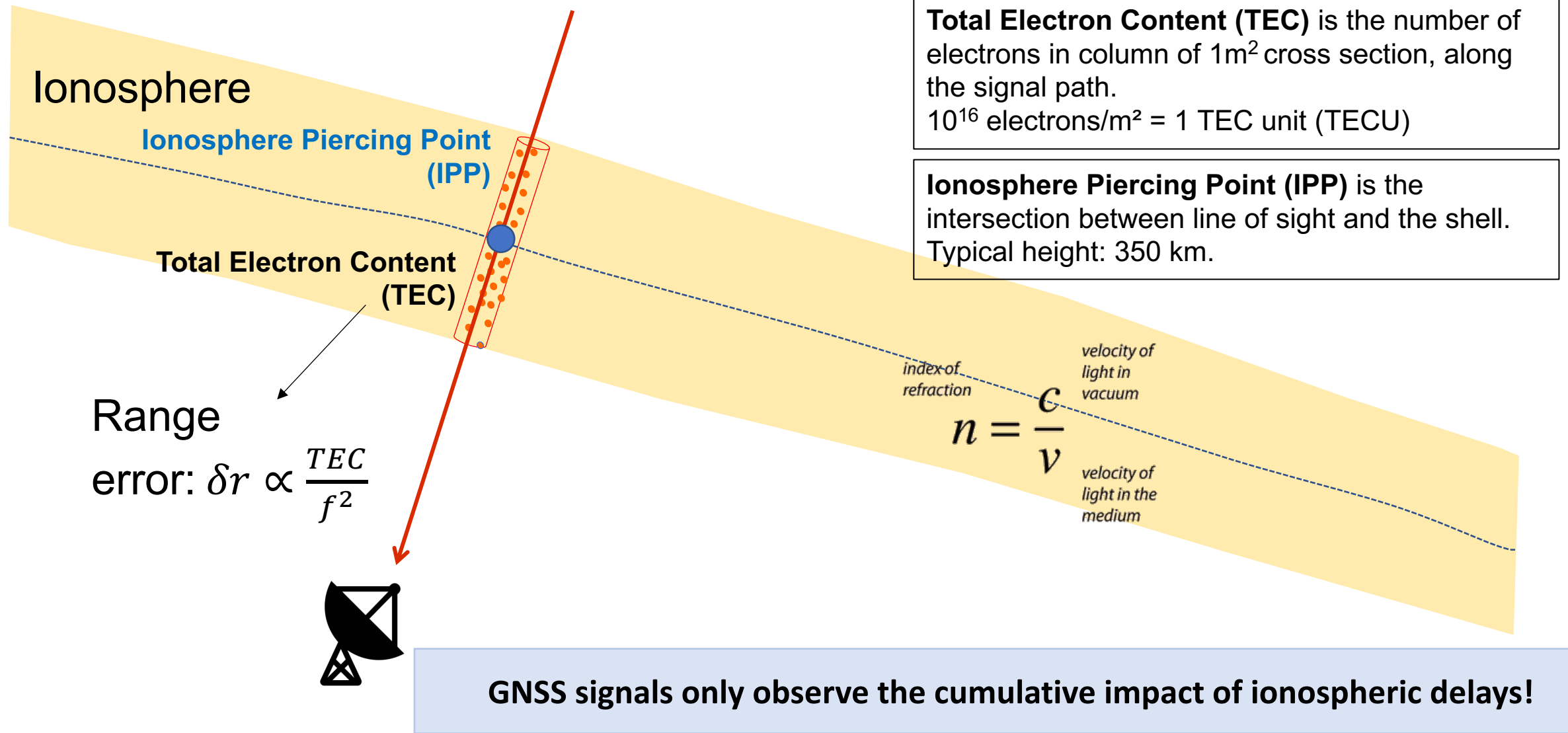
Credit: NASA's Goddard Space Flight Center/Mary Pat Hrybyk-Keith



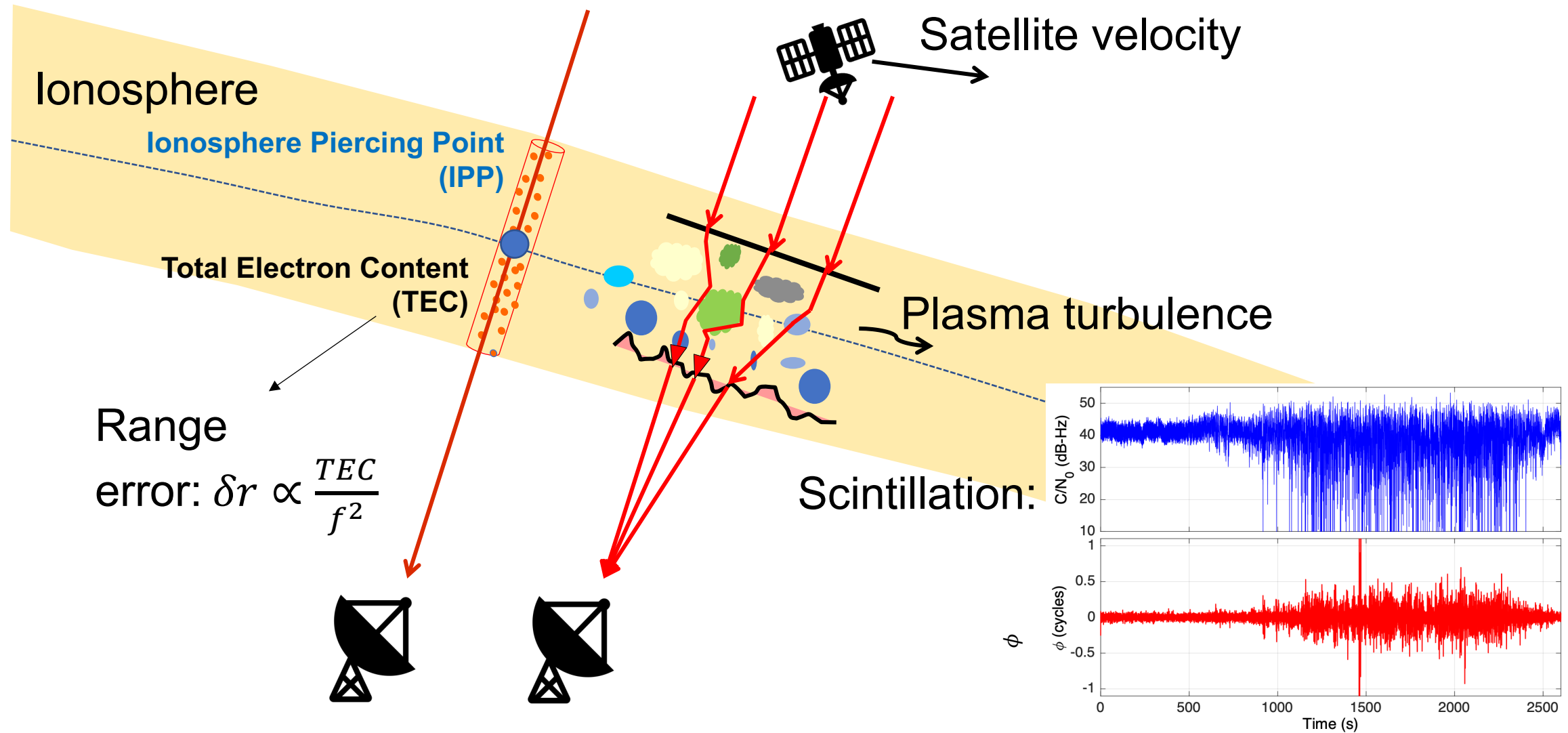
Outline of this lecture

- **What is it?** Introduction to ionosphere
- ➔ • **Why does it impact GNSS?** Group delay and phase advance
- **What can we do about it?** Ionospheric modeling, correction, and mapping
- **Ionospheric scintillation**
- **Other ionospheric effects**

Ionospheric TEC and Thin-Shell Model



Ionospheric Scintillation



Index of Refraction of a Cold Plasma: Appleton-Hartree Equation

$$n^2 = 1 - \frac{X}{1 - iZ - \frac{\frac{1}{2}Y^2 \sin^2 \theta}{1 - X - iZ} \pm \frac{1}{1 - X - iZ} \left(\frac{1}{4}Y^4 \sin^4 \theta + Y^2 \cos^2 \theta (1 - X - iZ)^2 \right)^{1/2}}$$

$$X = \left(\frac{f_p}{f} \right)^2 \quad f_p = \frac{1}{2\pi} \sqrt{\frac{e^2 N_e}{\epsilon_0 m_e}}$$

$$Y = \frac{f_g}{f} \quad f_g = \frac{|e|B}{2\pi m_e} \approx 28 \times 10^9 B \approx 1 \text{ MHz}$$

N_e : electron density

f_p : electron plasma frequency

f_g : electron gyro frequency

ϵ_0 : permittivity of free space

m_e : electron mass

e : electron charge

B : magnetic field strength

θ : angle between the magnetic field and wave vectors

Approximation via Taylor Expansion

$$n_\phi \approx 1 - \underbrace{\frac{X}{2}}_{\text{1st order}} \pm \underbrace{XY |\cos \theta_B|}_{\text{2nd order}} - \underbrace{\frac{1}{4}X \left(\frac{X}{2} + Y^2 (1 + \cos^2 \theta_B) \right)}_{\text{3rd order}}$$

Vacuum

$$n_\phi \approx 1 - \frac{X}{2} \approx 1 - \frac{e^2}{2m_e \epsilon_0 f^2} N_e = 1 - 40.3 \frac{N_e}{f^2}$$

Ionosphere is dispersive so its index of refraction is frequency dependent!

Group and Phase Velocity Relationship

- **Phase velocity**: constant carrier phase propagation speed
- **Group velocity**: constant modulating envelope propagation speed



By Kraaiennest - Own work, CC BY-SA 4.0, <https://commons.wikimedia.org/w/index.php?curid=3651297>

Ionospheric phase refractive index

$$n_p = \frac{c}{v_p}$$

$$n_p = 1 - \frac{40.3N_e}{f^2} \quad \longrightarrow \quad n_g = n_p + f \frac{dn_p}{df} \quad \longrightarrow \quad n_g = 1 + \frac{40.3N_e}{f^2}$$

Ionospheric group refractive index

$$n_g = \frac{c}{v_g}$$

Ionospheric group delay is equivalent to its phase advance

Ionosphere Refractive Index & GPS Signal Propagation

Define: Total Electron Content: $TEC = \int_S^R n_e(l) dl$

$$\Delta\tau = \tau - \tau_{vacuum} = \frac{1}{c} \int_S^R (n(l) - 1) dl$$

$$\Delta\tau_p = \frac{1}{c} \int_S^R (n_p - 1) dl = \frac{1}{c} \int_S^R \left(-\frac{40.3 n_e}{f^2} \right) dl = -\frac{40.3}{c f^2} TEC$$

$$I_p = c \Delta\tau_p = -\frac{40.3}{f^2} TEC$$

$$I_g = c \Delta\tau_g = \frac{40.3}{f^2} TEC = -I_p$$

Define 1 TEC unit = 10^{16} electrons/m², corresponds to $I_g = -I_p = 0.16$ m at GPS L1 frequency.



GPS Signal Ionospheric Effects

$$\rho(t) = r(t, t - \tau) + I_{\rho}(t) + T_{\rho}(t) + c(\delta t_u(t) - \delta t^s(t - \tau)) + \varepsilon_{\rho}(t)$$

$$\varphi(t) = \frac{c\tau}{\lambda} + N = \frac{1}{\lambda} \left(r(t, t - \tau) + I_{\varphi}(t) + T_{\varphi}(t) + c\delta t_u(t) - c\delta t^s(t - \tau) \right) + \varepsilon_{\varphi}(t) + N(t)$$

$$I_{\rho 1} = -I_{\varphi 1} = \frac{40.3 \text{ TEC}}{f_1^2}$$

99.9% accuracy, an approximation of the “first-order” terms



Outline of this lecture

- **What is it?** Introduction to ionosphere
- **Why does it impact GNSS?** Group delay and phase advance
- ➔ • **What can we do about it?** Ionospheric modeling, correction, and mapping
- **Ionospheric scintillation**
- **Other ionospheric effects**

Dual Frequency Receiver Estimation of Absolute TEC

Recall code measurement model: $\rho = r + I_\rho + T + c(\delta t_u - \delta t^s) + \varepsilon_\rho$

$$\text{L1: } \rho_1 = r + I_{\rho 1} + T + c(\delta t_u - \delta t^s) + \varepsilon_{\rho 1} = I_{\rho 1} + \rho_{1IF} \approx I_{\rho 1} + \rho_{IF} \quad I_{\rho 1} = \frac{40.3TEC}{f_1^2}$$

$$\text{L2: } \rho_2 = r + I_{\rho 2} + T + c(\delta t_u - \delta t^s) + \varepsilon_{\rho 2} = I_{\rho 2} + \rho_{\rho 2IF} \approx I_{\rho 2} + \rho_{IF} \quad I_{L2} = \frac{40.3TEC}{f_{L2}^2}$$

We can compute TEC based on ρ_1 and ρ_2 :

$$\rho_1 - \rho_2 = I_{\rho 1} - I_{\rho 2} = 40.3TEC \left(\frac{1}{f_1^2} - \frac{1}{f_2^2} \right) \Rightarrow TEC = \frac{f_1^2 f_2^2}{40.3(f_1^2 - f_2^2)} (\rho_2 - \rho_1)$$

Dual frequency pseudorange differences can isolate ionospheric delay



Ionosphere-free Pseudorange ρ_{IF} : ionosphere-free range

Recall code measurement model: $\rho = r + I_\rho + T + c(\delta t_u - \delta t^s) + \epsilon_\rho$

$$\text{L1: } \rho_1 = r + I_{\rho_1} + T + c(\delta t_u - \delta t^s) + \epsilon_{\rho_1} = I_{\rho_1} + \rho_{1IF} \approx I_{\rho_1} + \rho_{IF}$$

$$\text{L2: } \rho_2 = r + I_{\rho_2} + T + c(\delta t_u - \delta t^s) + \epsilon_{\rho_2} = I_{\rho_2} + \rho_{\rho_2IF} \approx I_{\rho_2} + \rho_{IF}$$

$$I_{\rho_1} = \frac{40.3TEC}{f_1^2}$$

$$I_{L2} = \frac{40.3TEC}{f_{L2}^2}$$

Ionosphere-free pseudoranges are more noisy than individual pseudoranges.

$$I_{\rho L1} = \frac{f_2^2}{f_1^2 - f_2^2} (\rho_{L2} - \rho_{L1}) \quad \text{Ionospheric delay on L1 pseudorange}$$

$$\rho_{IF} = \rho_{L3} = \frac{f_1^2}{f_1^2 - f_2^2} \rho_{L1} - \frac{f_2^2}{f_1^2 - f_2^2} \rho_{L2} \quad \text{Ionosphere-free pseudorange}$$

$$\rho_{IF} = 2.546\rho_{L1} - 1.546\rho_{L2} \quad \text{Approximate just to get an idea – don't use these numerical values}$$



Dual Frequency Receiver Estimation of Relative TEC

Recall carrier measurement model:

$$\phi = \frac{1}{\lambda} (r - I_{\rho} + T) + \frac{c}{\lambda} (\delta t_u - \delta t^s) + N + \varepsilon_{\phi}$$

$$\text{L1: } \phi_1 = -I_{\rho 1} + \phi_{IF} + N_1$$

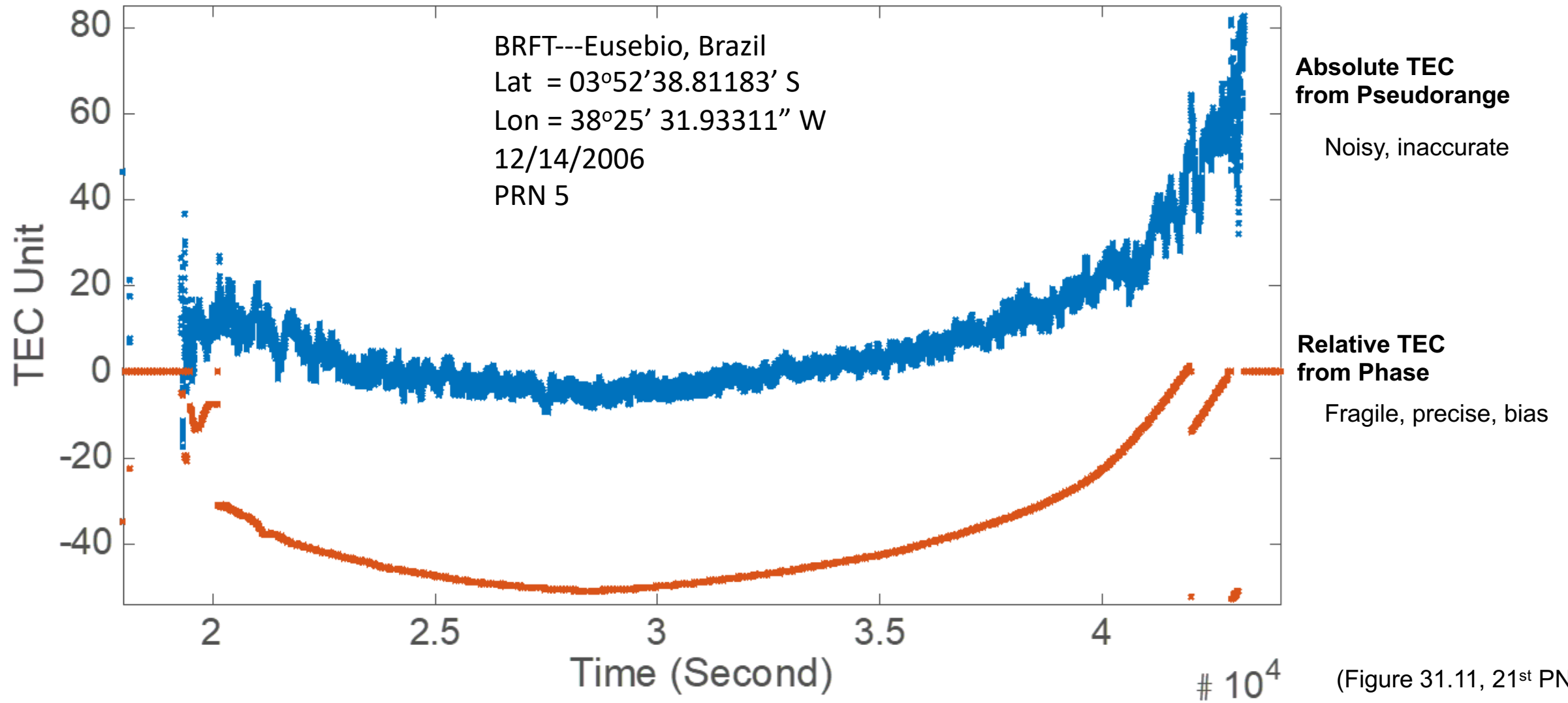
$$\text{L2: } \phi_2 = -I_{\rho 2} + \phi_{IF} + N_2$$

$$TEC = \frac{f_1^2 f_2^2}{40.3(f_1^2 - f_2^2)} (\lambda_1 (\phi_1 - N_1) - \lambda_2 (\phi_2 - N_2))$$

$$\text{Relative TEC: } \Delta TEC = \frac{f_1^2 f_2^2}{40.3(f_1^2 - f_2^2)} (\lambda_1 \phi_1 - \lambda_2 \phi_2)$$



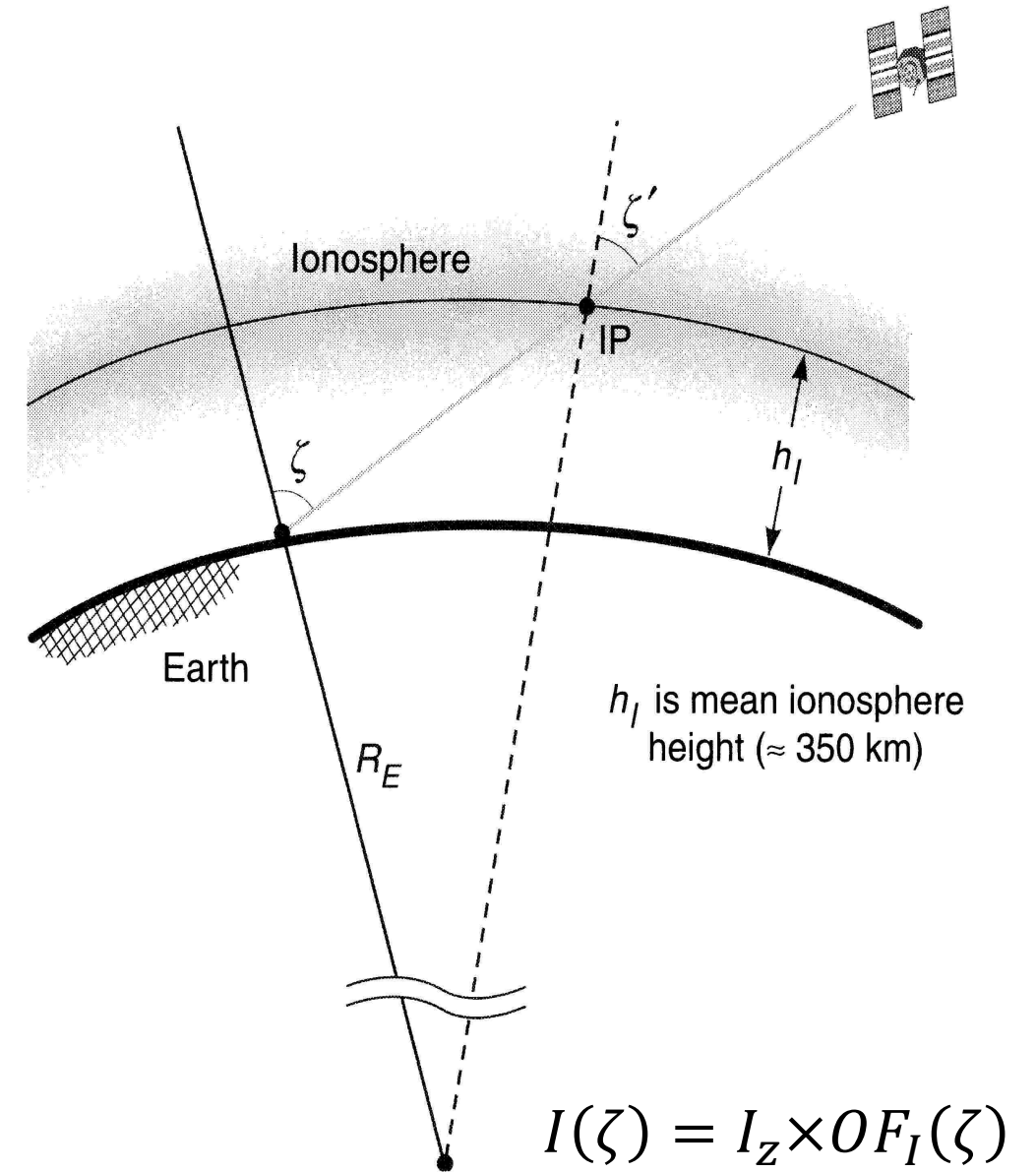
Example Dual Frequency Ionosphere TEC Measurements



(Figure 31.11, 21st PNT)

Ionospheric Zenith Delay

- The total or slant ionospheric delay depends on the length of the ray path through the ionosphere and how active the ionosphere is in the region it is passing through.
- To compare the activity levels the total delay can be approximated as the product of a zenith delay and a geometrical obliquity factor which depends on the satellite zenith angle (compliment of the elevation angle).
- Zenith delay is modeled as applying to an observer on the surface of the Earth directly below the IPP (ionospheric pierce point).
- GPS measurements at stations around the globe are used to observe VTEC.



Obliquity Factor (VTEC-to-STECh)

$$STECh(\zeta) = \frac{1}{\cos \zeta'} VTECh$$

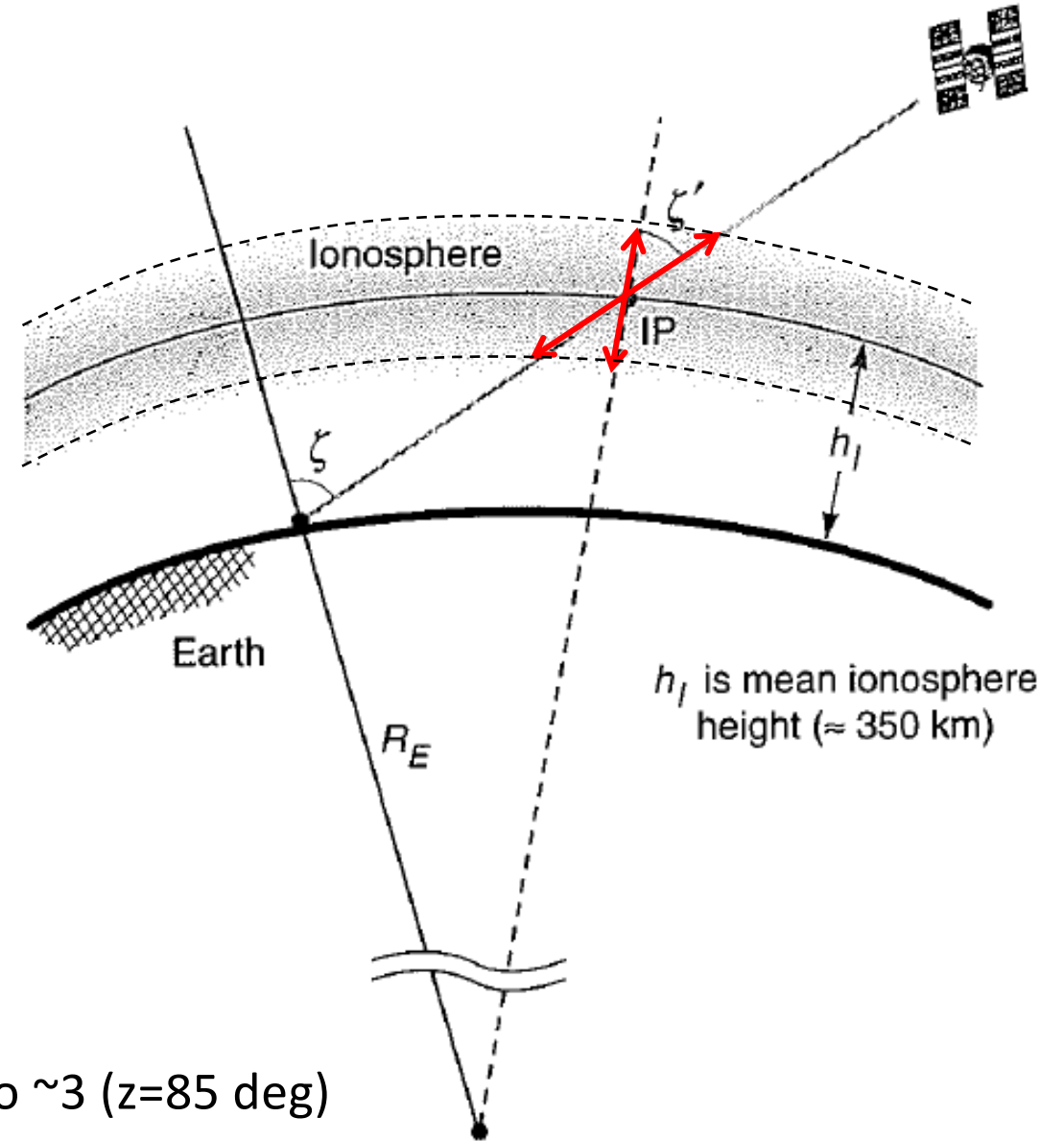
Law of sines: $\frac{\sin \zeta}{R_E + h_I} = \frac{\sin \zeta'}{R_E}$

$$OF_I = \frac{1}{\cos \zeta'} = \frac{1}{\sqrt{1 - \left(\frac{R_E \sin \zeta}{R_E + h_I}\right)^2}}$$

$$OF_I \approx 1.0 + 16.0 \times (0.53 - el)^3$$

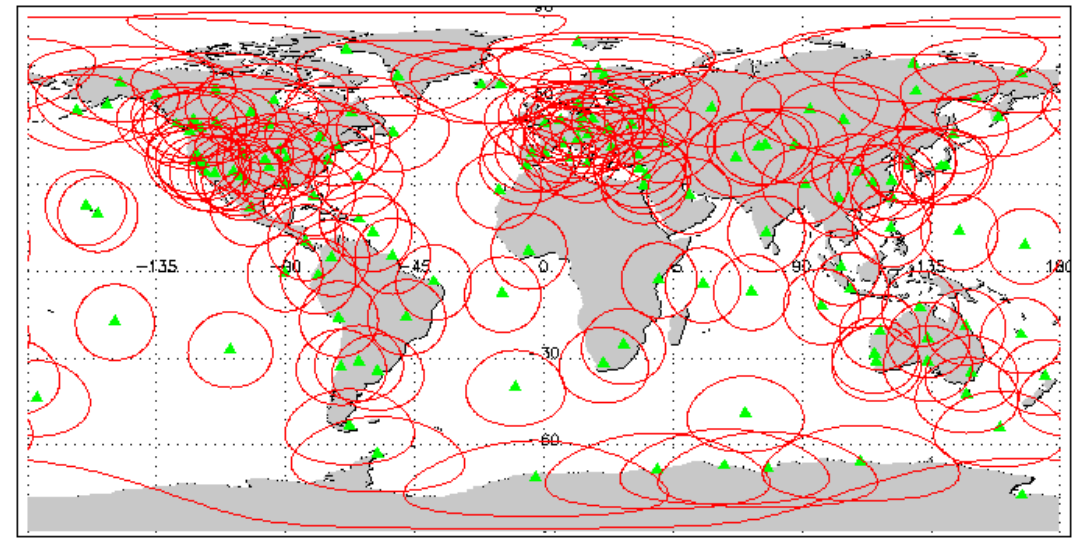
$$I(\zeta) = I_z \times OF_I(\zeta) \quad I_z: \text{Zenith delay}$$

OF_I ranges from 1 ($z=0$) to ~ 3 ($z=85$ deg)

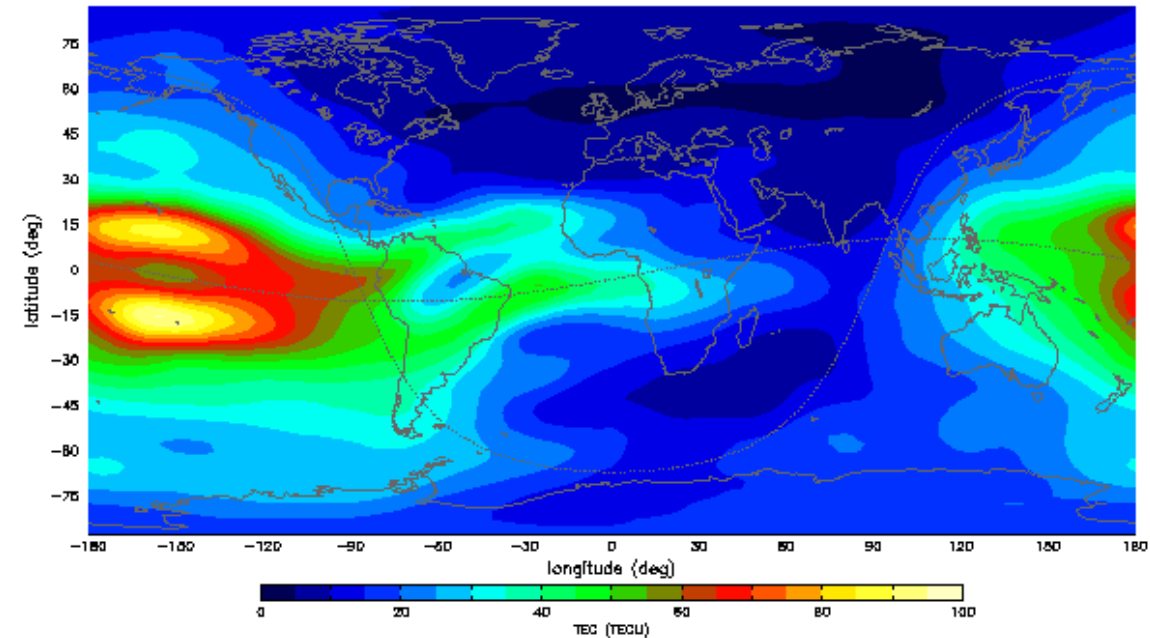


Global Ionospheric Maps

- Global Ionospheric Maps (GIMs) are generated using data from carefully sited GPS receivers (JPL network shown to the right).
- Generated by:
 - JPL (NASA, California)
 - Center for Orbit Determination (CODE, U. of Bern, Switzerland)
 - GSD, Canada
 - UPCG
 - Others
- Updated (at least) daily, used for GPS and research applications
- Calculates TECV from ranging error



TEC MAP (height= 450.0 km) at 2023/01/01,00:00:00
gAGE/UPC IONEX file containing the COMBINED IGS TEC MAPS and DCBs



Ionosphere Error Correction Broadcast (Klobuchar) Model

$$\frac{I_{L1,z}}{c} = \begin{cases} A_1 + A_2 \cos\left(\frac{2\pi(t - A_3)}{A_4}\right) & \text{if } |t - A_3| < \frac{A_4}{4} \\ A_1 & \text{otherwise} \end{cases}$$

t : Local time at IPP

$A_1 = 5 \times 10^{-9} \text{ s}$ Nighttime zenith delay

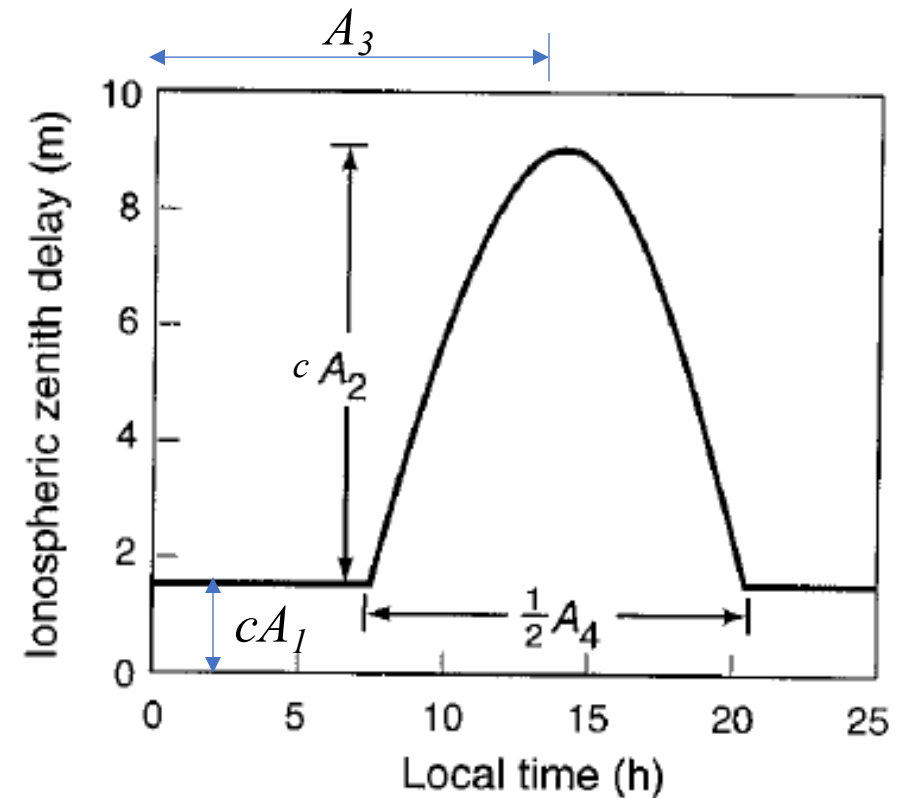
$A_3 = 50,400 \text{ s (LT)}$ Phase of the cosine peak

$$A_2 = \alpha_0 + \alpha_1 \Phi_m + \alpha_2 \Phi_m^2 + \alpha_3 \Phi_m^3$$

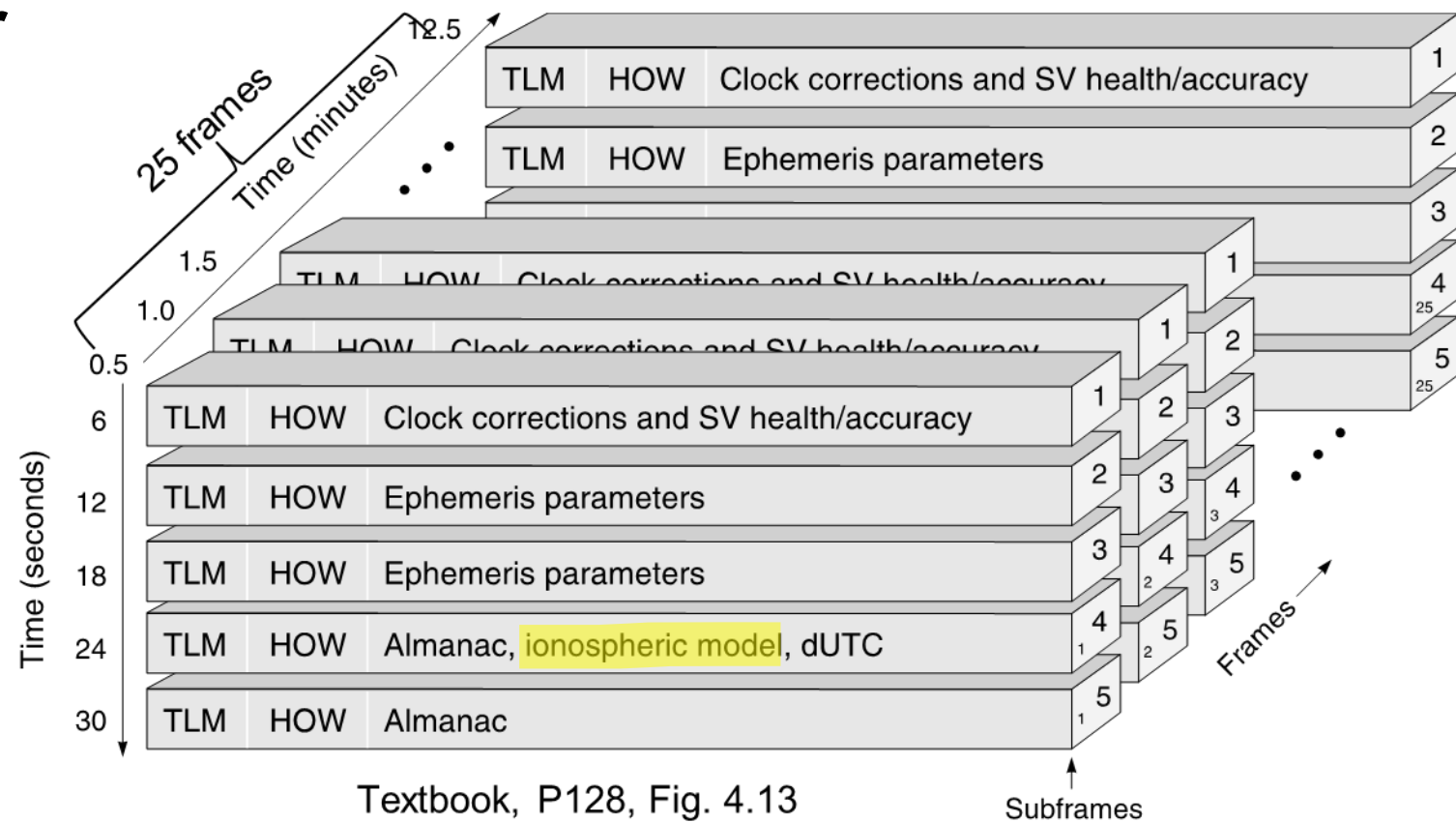
$$A_4 = \beta_0 + \beta_1 \Phi_m + \beta_2 \Phi_m^2 + \beta_3 \Phi_m^3$$

Φ_m : **IPP** geomagnetic latitude

$\alpha_0, \alpha_1, \alpha_2, \alpha_3$ Broadcast in the navigation message
 $\beta_0, \beta_1, \beta_2, \beta_3$ Updated once every 6 days
 Based on ~370 sensors measurements



Where is the Klobuchar Model stored?



2	NAVIGATION DATA	RINEX VERSION / TYPE
CCRINEXN V1.6.0 UX CDDIS	14-OCT-07 02:51	PGM / RUN BY / DATE
IGS BROADCAST EPHEMERIS FILE		COMMENT
0.1118E-07 0.0000E+00 -0.5960E-07 0.0000E+00		ION ALPHA
0.9011E+05 0.0000E+00 -0.1966E+06 0.0000E+00		ION BETA

NeQuick-G, BDGIM, and Comparison

NeQuick-G:

Semi-empirical model &
ESA released the NeQuick Code

$$A_z = a_0 + a_1\mu + a_2\mu^2 \quad (31.22)$$

where μ is the modified dip latitude (MODIP) and is determined by the true magnetic inclination I in the ionosphere and the geographic latitude of the receiver φ :

$$\tan \mu = \frac{I}{\sqrt{\cos \varphi}} \quad (31.23)$$

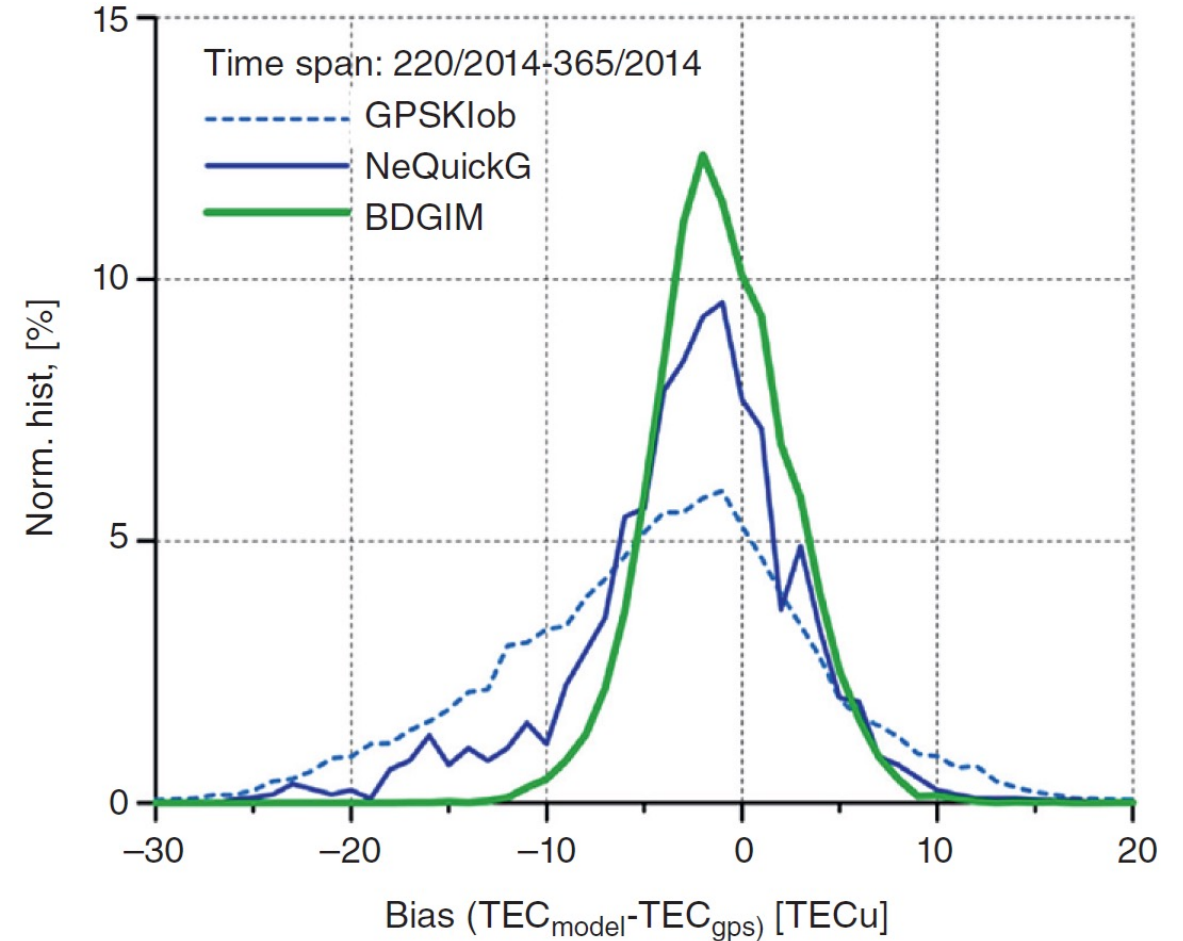
BDGIM:

A table of predicted & non-broadcast
coefficients (Table 31.1, Ch31)

$$VTEC = A_0 + \sum_{i=1}^9 \alpha_i A_i \quad (31.15)$$

The nine coefficients α_i ($i = 1$ to 9) are provided in the BDS navigation message. A_i can be computed using the following formula:

$$A_i = \begin{cases} \tilde{P}_{|n_i|,|m_i|}(\sin \varphi') \times \cos(m_i \lambda') & m_i \geq 0 \\ \tilde{P}_{|n_i|,|m_i|}(\sin \varphi') \times \sin(-m_i \lambda') & m_i < 0 \end{cases} \quad (31.16)$$

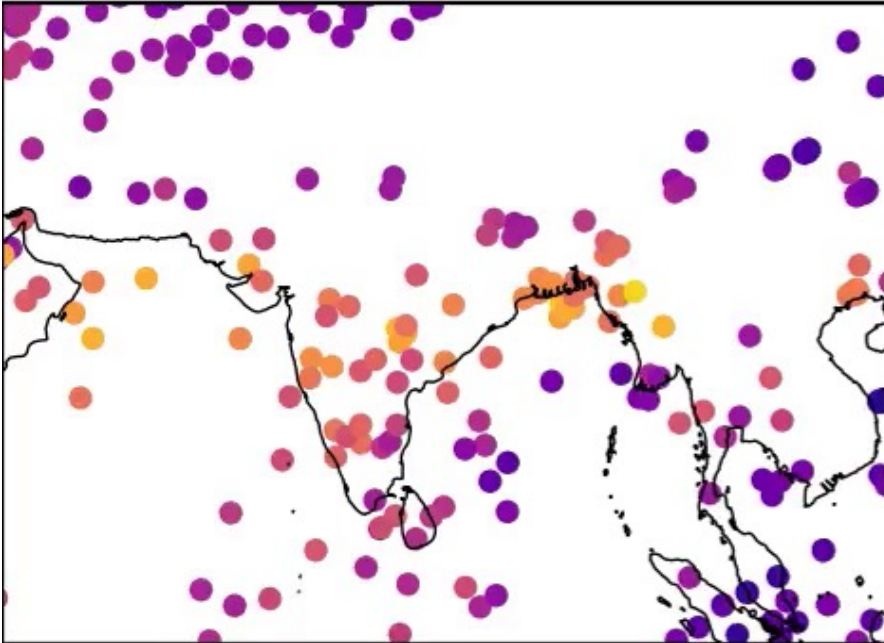


(21st Century PNT, Ch31)

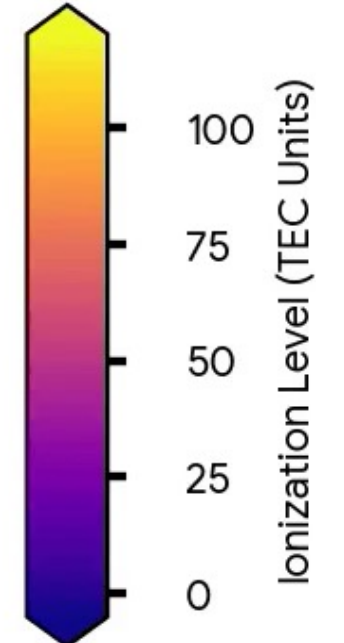
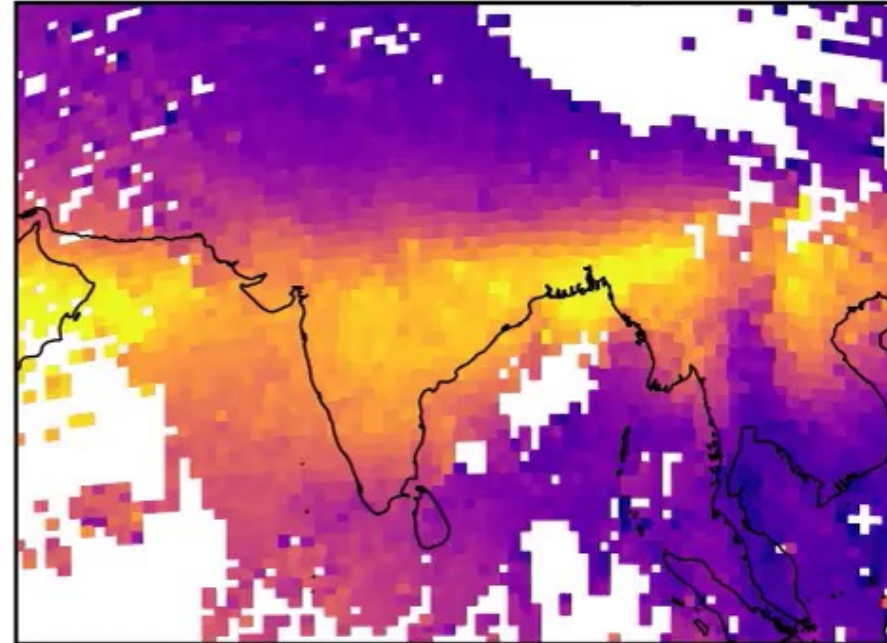


Global Ionospheric Maps from Smart Phone GNSS

Using Stations



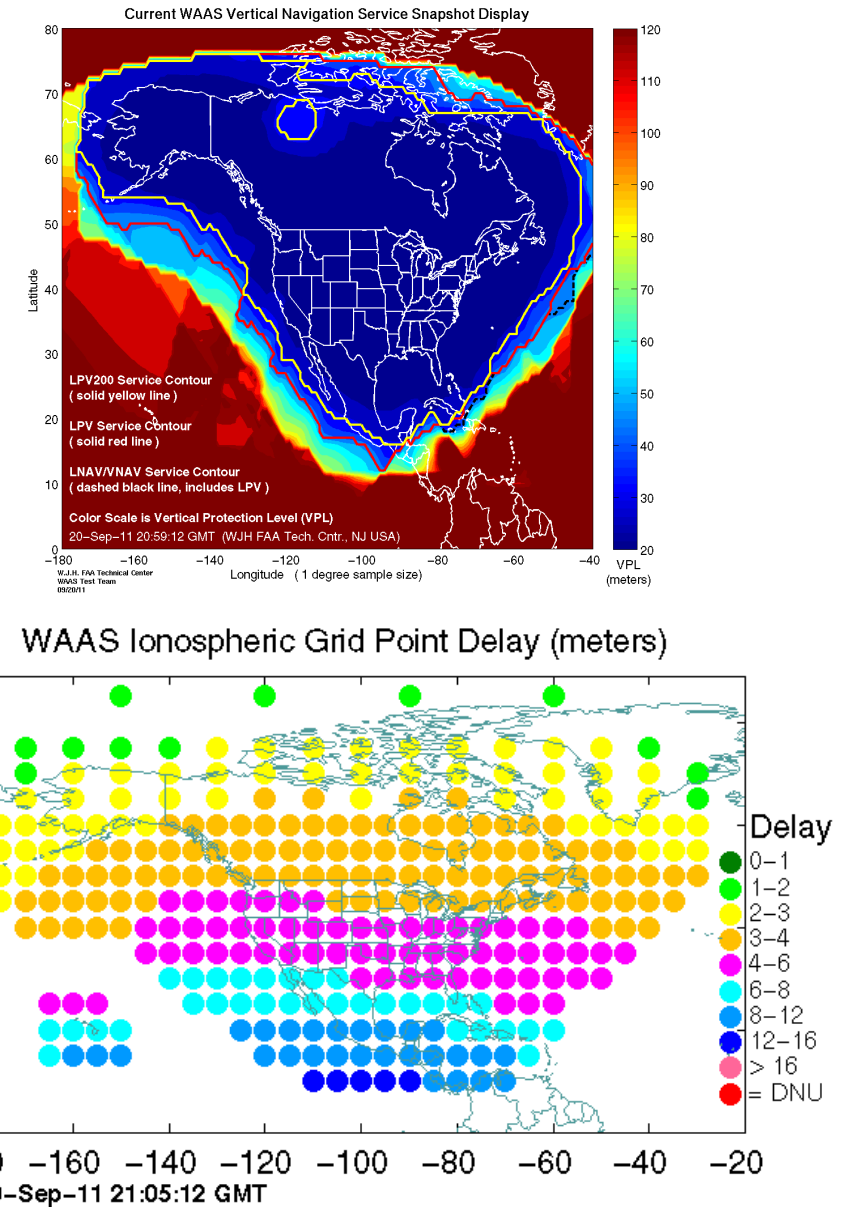
Using Phones



<https://research.google/blog/mapping-the-ionosphere-with-the-power-of-android/>

WAAS (Wide Area Augmentation System)

- ~90-95% correct; guaranteed 7.6 m, actual ~ 1 m accuracy
- WAAS supplies correction parameters for a number of points (organized in a grid pattern) across the WAAS service area.
- For each satellite observation the user receiver interpolates to its pierce point location based on the four closest grid points
- Receiver then calculates total delay based on measured delays, using interpolation



Images taken from www.nstb.tc.faa.gov (public domain)

Outline of this lecture

- **What is it?** Introduction to ionosphere
- **Why does it impact GNSS?** Group delay and phase advance
- **What can we do about it?** Ionospheric modeling, correction, and mapping
- ➔ • **Ionospheric scintillation**
- **Other ionospheric effects**

Ionospheric Scintillation

- Rapid variations in the phase and amplitude of a radio signal that has passed through the ionosphere
- Caused by small scale variations in the ionospheric plasma density
- Described by S4 intensity-scintillation index: standard deviation of the intensity normalized by the mean intensity.
- Primary influence on GPS is the potential for receiver tracking loops to lose lock on the signal.

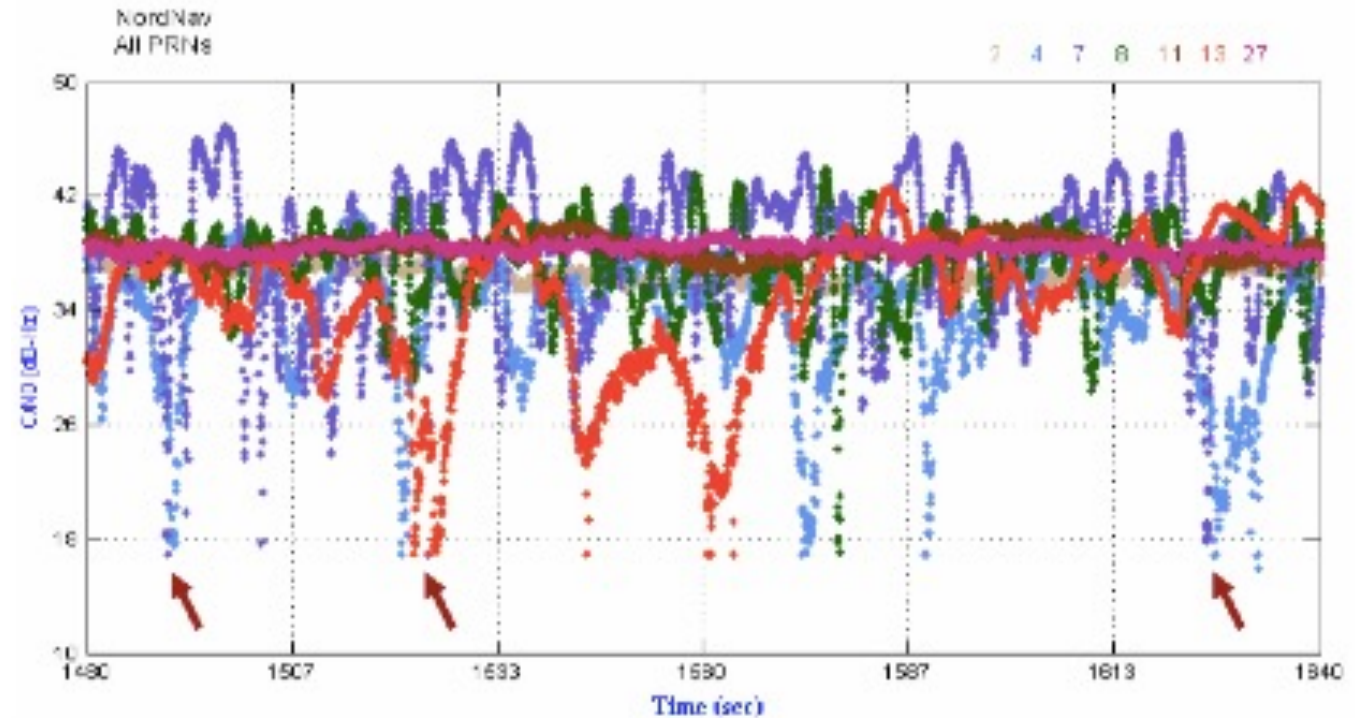
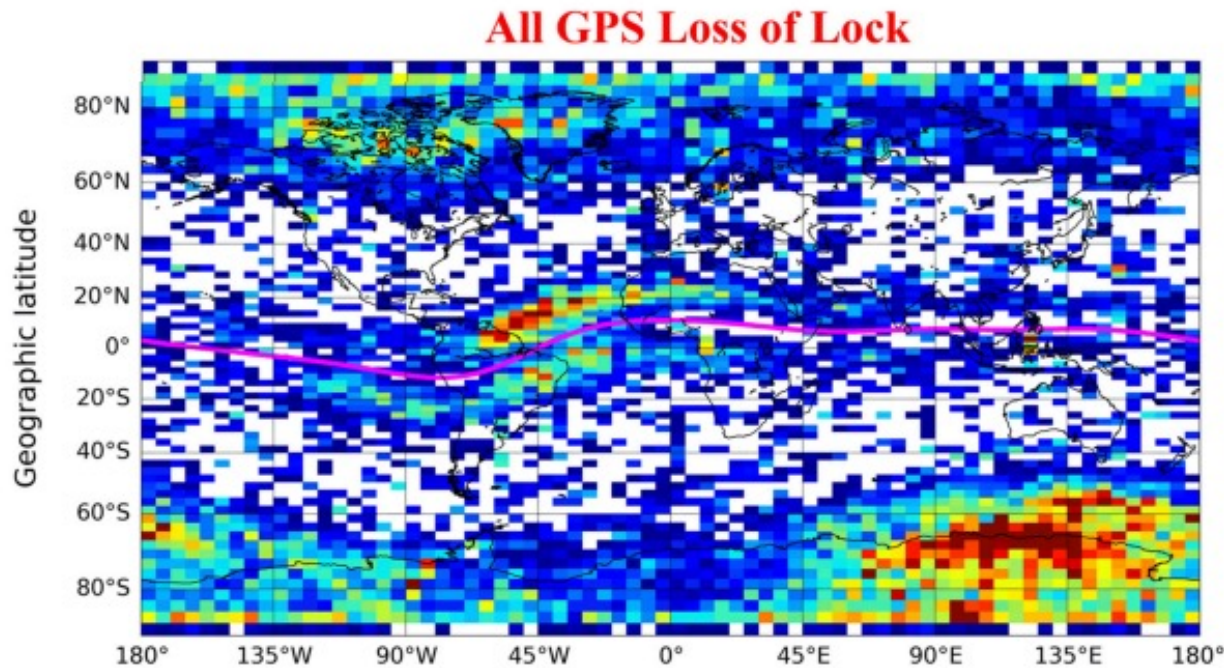


Figure 6. C/N₀ of All Seven Satellites during Solar Maximum

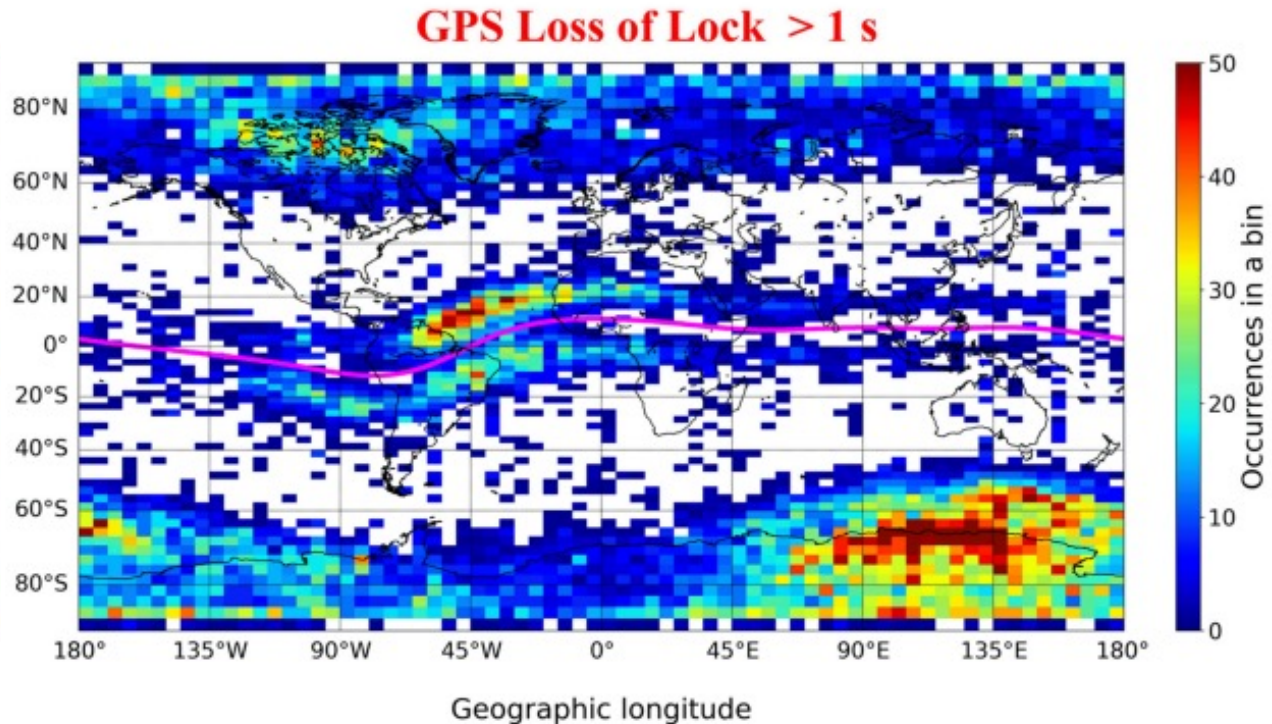
Seo, et al.

<http://waas.stanford.edu/~www/papers/gps/PDF/SeoIBSS07.pdf>

Ionospheric Scintillation Impacts



Swarm A and C form the lower pair of satellites flying side-by-side (1.4° separation in longitude at the equator) at an altitude of **462 km (initial altitude)** and at 87.35° inclination angle, whereas Swarm B is cruising at a higher orbit of **511 km (initial altitude)** and at 87.75° inclination angle.



Pezzopane, M., Pignalberi, A., Coco, I., Consolini, G., De Michelis, P., Giannattasio, F., ... & Tozzi, R. (2021). **Occurrence of GPS loss of lock based on a Swarm half-solar cycle dataset** and its relation to the background ionosphere. *Remote Sensing*, 13(11), 2209.

SUMMARY OF IONOSPHERIC EFFECTS

- Region of the Earth's atmosphere from ~ 100 to ~ 1000 km above the Earth's surface which includes active electrons
- Term describing it: total electron content (TEC).
- Depends on: Time of day (where the sun is), Season, Latitude, Solar (Sunspot) Cycle
- Delay on L1 (m) is approximately $40.3 \text{ TEC}/f_1^2$
- Corrected in the receiver by dual frequency measurements or Klobuchar model.
- Severe scintillation can cause receivers to lose lock.
Not a very common occurrence at mid latitudes.



Reading Assignment

- Misra & Enge Textbook Ch 5

Other References

- Position, Navigation, and Timing Technologies in the 21st Century, Ed by Y.T.J. Morton, et al., Ch 31 Ionospheric Effects, Monitoring, and Mitigation Techniques, Y. Jade Morton, Zhe Yang, Brian Breitsch, Harrison Bourne, Charles Rhino, 2021.
- 2017 Springer Handbook of Global Navigation Satellite Systems, Springer Handbooks, doi 10.1007/978-3-319-42928-1_32, Peter J.G. Teunissen and Oliver Montenbruck (eds.)
- Ch. 6 Atmospheric Signal Propagation – T. Hobiger and N. Jakowski